MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3340 CONCENTRATE THICKENING

DESIGN CALCULATION OF MISCELLANEOUS FOUNDATION

AS BUILT

MD-216-3340-EG-CV-CAL-1222

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	16-11-2023	AS BUILT	SHD	SGC	PRI	
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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Structural Column Foundation for Concentrate Thickening area of Plant.

2. SCOPE

The document is prepared for Analysis and Design of Structural Column Foundation for Concentrate Thickening area of Plant.
This includes foundation of Trestles for Conveyor gallery marked CNV-3360-002 as well as Vertical Gravity take Up Unit.
Also Column Foundations for Structure supporting the said gallery, Flotation Tanks & Stair Tower have been taken care in this document.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Foundations mkd. F2, F4, F5, F6, F7 & F9 are modelled in & analysed in Staad-Pro. . Support Reactions from Superstructure have been applied as loads to Foundations.
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008 (as supported by STAAD-Pro).
Net Safe Bearing Capacilty is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3. 3340 STAIR TOWER

3.1 Design of Foundation for Stair Tower marked F1

(Refer Calculation No. MD-216-3340-EG-ST-CAL-2192)

	1 (KN)			2 (KN)			3 (KN)			4 (KN)		
Load Detail	FX	FY	FZ	FX	FY	FZ	FX	FY	FZ	FX	FY	FZ
DEAD LOAD	0	28	1	0	34	0	-2	37	-1	2	28	0
LIVE LOAD	0	39	1	0	29	-1	-7	121	-1	6	149	1
WIND LOAD_X	-12	-58	0	-12	58	0	-11	56	0	-11	-56	0
WIND LOAD_Z	0	-27	-11	0	-25	-10	0	25	-10	0	27	-11

STRENGTH DESIGN			
Load Combinations	FX (T)	FY (T)	FZ (T)
SUPPORT 1			
1.4 DL	0.00	3.92	0.14
1.4 DL + 1.7 LL	0.0	10.6	0.3
1.2 DL + 1.6 WLx + LL	-1.9	-2.0	0.2
1.2 DL + 1.6 WLz + LL	0.0	2.9	-1.5
0.9 DL + 1.6 WLx	-1.9	-6.8	0.1
0.9 DL + 1.6 WLz	0.0	-1.8	-1.7
SUPPORT 2			
1.4 DL	0.0	4.8	0.0
1.4 DL + 1.7 LL	0.0	9.7	-0.2
1.2 DL + 1.6 WLx + LL	-1.9	16.3	-0.1
1.2 DL + 1.6 WLz + LL	0.0	3.0	-1.7
0.9 DL + 1.6 WLx	-1.9	12.3	0.0
0.9 DL + 1.6 WLz	0.0	-0.9	-1.6
SUPPORT 3			
1.4 DL	-0.3	5.2	-0.1
1.4 DL + 1.7 LL	-1.5	25.8	-0.3
1.2 DL + 1.6 WLx + LL	-2.7	25.5	-0.2
1.2 DL + 1.6 WLz + LL	-0.9	20.5	-1.8
0.9 DL + 1.6 WLx	-1.9	12.3	-0.1
0.9 DL + 1.6 WLz	-0.2	7.3	-1.7
SUPPORT 4			
1.4 DL	0.3	3.9	0.0
1.4 DL + 1.7 LL	1.3	29.3	0.2
1.2 DL + 1.6 WLx + LL	-0.9	9.3	0.1
1.2 DL + 1.6 WLz + LL	0.8	22.6	-1.7
0.9 DL + 1.6 WLx	-1.6	-6.4	0.0
0.9 DL + 1.6 WLz	0.2	6.8	-1.8

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DESIGN OF FOUNDATION STAIR TOWER

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.85m	1.85m
Thickness of raft :	0.25 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	WL

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.85*1.85*0.25*2.5)+(0.6*0.6*1.75*2.5) = 3.71T

Weight of soil = ((1.85*1.85)-(0.6*0.6))*1.25*1.8 = 6.89T

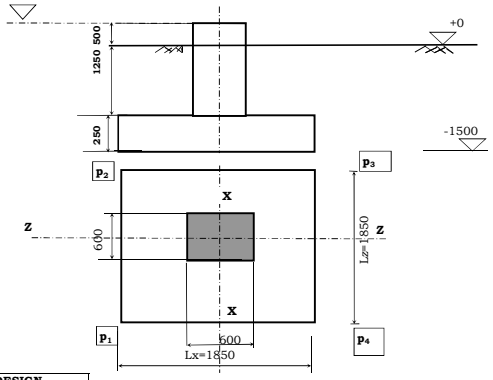
U/S Base Pl Total = 10.60T

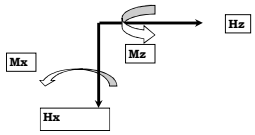
Effective depth of slab along X direction = 169mm = 0.169m

Effective depth of slab along Z direction = 157mm = 0.157m

Clear projection along Z direction = 0.625m

Clear projection along X direction = 0.625m





WORKING STRESS DESIGN

Load Case	Load at U/S Of Base Plt					Load at bottom of foundation				
SUPPORT 1	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	6.70	0.20	0.0	0.00	17.30	0.00	0.40	0.20	0.00
DL+0.75*LL+0.75*WLx	-0.90	1.38	0.18	0.0	0.00	11.98	-0.90	0.35	0.18	1.80
DL+0.75*LL+0.75*WLz	0.00	3.70	-0.65	0.0	0.00	14.30	0.00	1.30	-0.65	0.00
0.6*DL+WLx	-1.20	-5.56	0.01	0.0	0.00	5.04	-1.20	0.02	0.01	2.40
0.6*DL+WLz	0.00	1.18	-0.56	0.0	0.00	11.78	0.00	1.12	-0.56	0.00
SUPPORT 2										
DL+LL	0.00	6.30	-0.10	0.0	0.00	16.90	0.00	0.20	-0.10	0.00
DL+0.75*LL+0.75*WLx	-0.90	9.93	-0.08	0.0	0.00	20.53	-0.90	0.15	-0.08	1.80
DL+0.75*LL+0.75*WLz	0.00	3.70	-0.83	0.0	0.00	14.30	0.00	1.65	-0.83	0.00
0.6*DL+WLx	-1.20	6.09	0.00	0.0	0.00	16.69	-1.20	0.00	0.00	2.40
0.6*DL+WLz	0.00	1.90	-0.60	0.0	0.00	12.50	0.00	1.20	-0.60	0.00
SUPPORT 3										
DL+LL	-0.90	15.80	-0.20	0.0	0.00	26.40	-0.90	0.40	-0.20	1.80
DL+0.75*LL+0.75*WLx	-1.55	16.98	-0.18	0.0	0.00	27.58	-1.55	0.35	-0.18	3.10
DL+0.75*LL+0.75*WLz	-0.73	14.65	-0.93	0.0	0.00	25.25	-0.73	1.85	-0.93	1.45
0.6*DL+WLx	-1.12	5.91	-0.01	0.0	0.00	16.51	-1.12	0.02	-0.01	2.23
0.6*DL+WLz	-0.20	5.20	-0.70	0.0	0.00	15.80	-0.20	1.40	-0.70	0.40
SUPPORT 4										
DL+LL	0.80	17.70	0.10	0.0	0.00	28.30	0.80	0.20	0.10	1.60
DL+0.75*LL+0.75*WLx	-0.18	9.78	0.08	0.0	0.00	20.38	-0.18	0.15	0.08	0.35
DL+0.75*LL+0.75*WLz	0.65	16.00	-0.75	0.0	0.00	26.60	0.65	1.50	-0.75	1.30
0.6*DL+WLx	-1.08	-5.36	0.00	0.0	0.00	5.24	-1.08	0.00	0.00	2.17
0.6*DL+WLz	0.20	4.42	-0.66	0.0	0.00	15.02	0.20	1.32	-0.66	0.40

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CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
SUPPORT 1									
DL+LL	17.30	0.00	0.00	0.20	0.40	NA	40.01	NA	34.60
DL+0.75*LL+0.75*WLx	11.98	-0.90	1.80	0.18	0.35	6.15	31.65	5.32	27.37
DL+0.75*LL+0.75*WLz	14.30	0.00	0.00	-0.65	1.30	NA	10.18	NA	8.80
0.6*DL+WLx	5.04	-1.20	2.40	0.01	0.02	1.94	277.24	1.68	239.77
0.6*DL+WLz	11.78	0.00	0.00	-0.56	1.12	NA	9.73	NA	8.41
SUPPORT 2									
DL+LL	16.90	0.00	0.00	-0.10	0.20	NA	78.16	NA	67.60
DL+0.75*LL+0.75*WLx	20.53	-0.90	1.80	-0.08	0.15	10.55	126.57	9.12	109.47
DL+0.75*LL+0.75*WLz	14.30	0.00	0.00	-0.83	1.65	NA	8.02	NA	6.93
0.6*DL+WLx	16.69	-1.20	2.40	0.00	0.00	6.43	NA	5.56	NA
0.6*DL+WLz	12.50	0.00	0.00	-0.60	1.20	NA	9.64	NA	8.33
SUPPORT 3									
DL+LL	26.40	-0.90	1.80	-0.20	0.40	13.57	61.05	11.73	52.80
DL+0.75*LL+0.75*WLx	27.58	-1.55	3.10	-0.18	0.35	8.23	72.88	7.12	63.03
DL+0.75*LL+0.75*WLz	25.25	-0.73	1.45	-0.93	1.85	16.11	12.63	13.93	10.92
0.6*DL+WLx	16.51	-1.12	2.23	-0.01	0.02	6.84	909.08	5.91	786.23
0.6*DL+WLz	15.80	-0.20	0.40	-0.70	1.40	36.54	10.44	31.60	9.03
SUPPORT 4									
DL+LL	28.30	0.80	1.60	0.10	0.20	16.36	130.89	14.15	113.20
DL+0.75*LL+0.75*WLx	20.38	-0.18	0.35	0.08	0.15	53.85	125.65	46.57	108.67
DL+0.75*LL+0.75*WLz	26.60	0.65	1.30	-0.75	1.50	18.93	16.40	16.37	14.19
0.6*DL+WLx	5.24	-1.08	2.17	0.00	0.00	2.24	NA	1.93	NA
0.6*DL+WLz	15.02	0.20	0.40	-0.66	1.32	34.73	10.53	30.04	9.10

Factor of safety against both Overturning & Sliding is 1.5. Hence it is OK.

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ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
SUPPORT 1	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4 DL	0.00	3.92	0.14	0.0	0.00	14.520	0.28	0.00
1.4 DL + 1.7 LL	0.00	10.55	0.31	0.0	0.00	21.150	0.62	0.00
1.2 DL + 1.6 WLx + LL	-1.92	-2.02	0.22	0.0	0.00	8.580	0.44	3.84
1.2 DL + 1.6 WLz + LL	0.00	2.94	-1.54	0.0	0.00	13.540	3.08	0.00
0.9 DL + 1.6 WLx	-1.92	-6.76	0.09	0.0	0.00	3.840	0.18	3.84
0.9 DL + 1.6 WLz	0.00	-1.80	-1.67	0.0	0.00	8.800	3.34	0.00
SUPPORT 2								
1.4 DL	0.00	4.76	0.00	0.0	0.00	15.360	0.00	0.00
1.4 DL + 1.7 LL	0.00	9.69	-0.17	0.0	0.00	20.290	0.34	0.00
1.2 DL + 1.6 WLx + LL	-1.92	16.26	-0.10	0.0	0.00	26.860	0.20	3.84
1.2 DL + 1.6 WLz + LL	0.00	2.98	-1.70	0.0	0.00	13.580	3.40	0.00
0.9 DL + 1.6 WLx	-1.92	12.34	0.00	0.0	0.00	22.940	0.00	3.84
0.9 DL + 1.6 WLz	0.00	-0.94	-1.60	0.0	0.00	9.660	3.20	0.00
SUPPORT 3								
1.4 DL	-0.28	5.18	-0.14	0.0	0.00	15.780	0.28	0.56
1.4 DL + 1.7 LL	-1.47	25.75	-0.31	0.0	0.00	36.350	0.62	2.94
1.2 DL + 1.6 WLx + LL	-2.70	25.50	-0.22	0.0	0.00	36.100	0.44	5.40
1.2 DL + 1.6 WLz + LL	-0.94	20.54	-1.82	0.0	0.00	31.140	3.64	1.88
0.9 DL + 1.6 WLx	-1.94	12.29	-0.09	0.0	0.00	22.890	0.18	3.88
0.9 DL + 1.6 WLz	-0.18	7.33	-1.69	0.0	0.00	17.930	3.38	0.36
SUPPORT 4								
1.4 DL	0.28	3.92	0.00	0.0	0.00	14.520	0.00	0.56
1.4 DL + 1.7 LL	1.30	29.25	0.17	0.0	0.00	39.850	0.34	2.60
1.2 DL + 1.6 WLx + LL	-0.92	9.30	0.10	0.0	0.00	19.900	0.20	1.84
1.2 DL + 1.6 WLz + LL	0.84	22.58	-1.66	0.0	0.00	33.180	3.32	1.68
0.9 DL + 1.6 WLx	-1.58	-6.44	0.00	0.0	0.00	4.160	0.00	3.16
0.9 DL + 1.6 WLz	0.18	6.84	-1.76	0.0	0.00	17.440	3.52	0.36

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

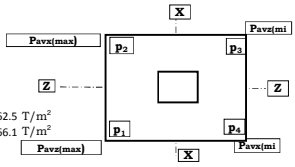
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² (BH28) For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



SUPPORT 1	Design Load/moments			Base Pressure (T/m²)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	17.30	0.00	0.40	5.05	0.00	0.38	7.43	6.68	7.43	6.68	5.18	O.K
DL+0.75*LL+0.75*WLx	11.98	1.80	0.35	3.50	1.71	0.33	7.54	6.87	4.12	3.46	5.29	O.K
DL+0.75*LL+0.75*WLz	14.30	0.00	1.30	4.18	0.00	1.23	7.41	4.95	7.41	4.95	5.16	O.K
0.6*DL+WLx	5.04	2.40	0.02	1.47	2.27	0.02	5.76	5.73	1.21	1.18	3.51	O.K
0.6*DL+WLz	11.78	0.00	1.12	3.44	0.00	1.06	6.50	4.38	6.50	4.38	4.25	O.K
SUPPORT 2												
DL+LL	16.90	0.00	0.20	4.94	0.00	0.19	7.13	6.75	7.13	6.75	4.88	O.K
DL+0.75*LL+0.75*WLx	20.53	1.80	0.15	6.00	1.71	0.14	9.84	9.56	6.43	6.15	7.59	O.K
DL+0.75*LL+0.75*WLz	14.30	0.00	1.65	4.18	0.00	1.56	7.74	4.61	7.74	4.61	5.49	O.K
0.6*DL+WLx	16.69	2.40	0.00	4.88	2.27	0.00	9.15	9.15	4.60	4.60	6.90	O.K
0.6*DL+WLz	12.50	0.00	1.20	3.65	0.00	1.14	6.79	4.52	6.79	4.52	4.54	O.K
SUPPORT 3												
DL+LL	26.40	1.80	0.40	7.71	1.71	0.38	11.80	11.04	8.39	7.63	9.55	O.K
DL+0.75*LL+0.75*WLx	27.58	3.10	0.35	8.06	2.94	0.33	13.33	12.66	7.45	6.79	11.08	O.K
DL+0.75*LL+0.75*WLz	25.25	1.45	1.85	7.38	1.37	1.75	12.50	9.00	9.76	6.25	10.25	O.K
0.6*DL+WLx	16.51	2.23	0.02	4.82	2.12	0.02	8.96	8.92	4.72	4.69	6.71	O.K
0.6*DL+WLz	15.80	0.40	1.40	4.62	0.38	1.33	8.32	5.67	7.56	4.91	6.07	O.K
SUPPORT 4												
DL+LL	28.30	1.60	0.20	8.27	1.52	0.19	11.97	11.60	8.94	8.56	9.72	O.K
DL+0.75*LL+0.75*WLx	20.38	0.35	0.15	5.95	0.33	0.14	8.43	8.14	7.76	7.48	6.18	O.K
DL+0.75*LL+0.75*WLz	26.60	1.30	1.50	7.77	1.23	1.42	12.43	9.58	9.96	7.12	10.18	O.K
0.6*DL+WLx	5.24	2.17	0.00	1.53	2.05	0.00	5.58	5.58	1.48	1.48	3.33	O.K
0.6*DL+WLz	15.02	0.40	1.32	4.39	0.38	1.25	8.02	5.52	7.26	4.76	5.77	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

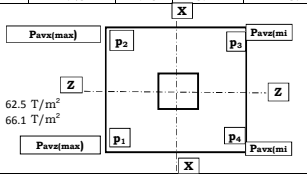
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisic	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

	Design Load/moments										
SUPPORT 1	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4 DL	14.5	0.00	0.28	4.24	0.00	0.27	6.51	5.98	6.51	5.98	4.26
1.4 DL + 1.7 LL	21.2	0.00	0.62	6.18	0.00	0.59	8.77	5.59	6.77	5.59	6.52
1.2 DL + 1.6 WLx + LL	8.6	3.84	0.44	2.51	3.64	0.42	8.56	5.73	-0.71	-1.55	6.31
1.2 DL + 1.6 WLz + LL	13.5	0.00	3.08	3.96	0.00	2.92	8.87	1.04	6.87	1.04	6.62
0.9 DL + 1.6 WLx	3.8	3.84	0.18	1.12	3.64	0.17	6.93	4.59	-2.35	-2.69	4.68
0.9 DL + 1.6 WLz	8.8	0.00	3.34	2.57	0.00	3.17	7.74	-0.59	5.74	-0.59	5.49
SUPPORT 2											
1.4 DL	15.4	0.00	0.00	4.49	0.00	0.00	6.49	4.49	4.49	4.49	4.24
1.4 DL + 1.7 LL	20.3	0.00	0.34	5.93	0.00	0.32	8.25	5.61	6.25	5.61	6.00
1.2 DL + 1.6 WLx + LL	26.9	3.84	0.20	7.85	3.64	0.19	13.68	11.30	4.40	4.02	11.43
1.2 DL + 1.6 WLz + LL	13.6	0.00	3.40	3.97	0.00	3.22	9.19	0.75	7.19	0.75	6.94
0.9 DL + 1.6 WLx	22.9	3.84	0.00	6.70	3.64	0.00	12.34	10.34	3.06	3.06	10.09
0.9 DL + 1.6 WLz	9.7	0.00	3.20	2.82	0.00	3.03	7.85	-0.21	5.85	-0.21	5.60
SUPPORT 3											
1.4 DL	15.8	0.56	0.28	4.61	0.53	0.27	7.41	4.88	4.35	3.81	5.16
1.4 DL + 1.7 LL	36.4	2.94	0.62	10.62	2.79	0.59	15.99	12.82	8.42	7.25	13.74
1.2 DL + 1.6 WLx + LL	36.1	5.40	0.44	10.55	5.12	0.42	18.08	15.25	5.85	5.01	15.83
1.2 DL + 1.6 WLz + LL	31.1	1.88	3.64	9.10	1.78	3.45	16.33	7.43	10.77	3.87	14.08
0.9 DL + 1.6 WLx	22.9	3.88	0.18	6.69	3.68	0.17	12.54	10.19	3.18	2.84	10.29
0.9 DL + 1.6 WLz	17.9	0.36	3.38	5.24	0.34	3.20	10.78	2.38	8.10	1.69	8.53
SUPPORT 4											
1.4 DL	14.5	0.56	0.00	4.24	0.53	0.00	6.77	4.77	3.71	3.71	4.52
1.4 DL + 1.7 LL	39.9	2.60	0.34	11.64	2.46	0.32	16.43	13.79	9.50	8.86	14.18
1.2 DL + 1.6 WLx + LL	19.9	1.84	0.20	5.81	1.74	0.19	9.75	7.37	4.26	3.88	7.50
1.2 DL + 1.6 WLz + LL	33.2	1.68	3.32	9.69	1.59	3.15	16.43	8.14	11.25	4.96	14.18
0.9 DL + 1.6 WLx	4.2	3.16	0.00	1.22	2.99	0.00	6.21	4.21	-1.78	-1.78	3.96
0.9 DL + 1.6 WLz	17.4	0.36	3.52	5.10	0.34	3.34	10.77	2.10	8.09	1.42	8.52

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

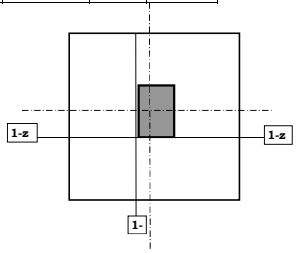
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.6+2*0.5*0.157)+(0.6+2*0.5*0.157))*2

= 3.028 m

Punching Area= ((0.6+2*0.5*0.157)*(0.6+2*0.5*0.157))

= 0.57 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.25*2.5+1.25*1.8)

= 2.88 T/m^2

Area of footing, A_f= 3.42 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 45.16 T

Strength of Concrete in shear Vc1 = 101.67 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 95.48 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 93.75 T **Table 22.6.5.2_ (c)**

Design Vc= 93.75 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (45.16*10000)/(3028*157)

= 0.95 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
SUPPORT 1												
1.4 DL	4.26	0.83	1.94	0.83	1.99	12.15	0.26	OK	0.74	0.127	OK	0.34
1.4 DL + 1.7 LL	6.52	1.27	2.97	1.27	3.05	18.60	0.39	OK	0.74	0.194	OK	0.52
1.2 DL + 1.6 WLx + LL	6.31	1.23	2.88	1.23	2.95	18.00	0.38	OK	0.74	0.188	OK	0.50
1.2 DL + 1.6 WLz + LL	6.62	1.29	3.02	1.29	3.10	18.88	0.4	OK	0.74	0.197	OK	0.52
0.9 DL + 1.6 WLx	4.68	0.91	2.13	0.91	2.19	13.35	0.28	OK	0.74	0.14	OK	0.37
0.9 DL + 1.6 WLz	5.49	1.07	2.50	1.07	2.57	15.66	0.33	OK	0.74	0.164	OK	0.44
SUPPORT 2												
1.4 DL	4.24	0.83	1.93	0.83	1.98	12.09	0.25	OK	0.74	0.126	OK	0.34
1.4 DL + 1.7 LL	6.00	1.17	2.74	1.17	2.81	17.12	0.36	OK	0.74	0.179	OK	0.48
1.2 DL + 1.6 WLx + LL	11.43	2.23	5.21	2.23	5.35	32.60	0.69	OK	0.74	0.341	OK	0.91
1.2 DL + 1.6 WLz + LL	6.94	1.36	3.16	1.36	3.25	19.80	0.42	OK	0.74	0.207	OK	0.55
0.9 DL + 1.6 WLx	10.09	1.97	4.60	1.97	4.72	28.78	0.61	OK	0.74	0.301	OK	0.80
0.9 DL + 1.6 WLz	5.60	1.09	2.55	1.09	2.62	15.97	0.34	OK	0.74	0.167	OK	0.44
SUPPORT 3												
1.4 DL	5.16	1.01	2.35	1.01	2.41	14.72	0.31	OK	0.74	0.154	OK	0.41
1.4 DL + 1.7 LL	13.74	2.68	6.27	2.68	6.43	39.19	0.82	OK	0.74	0.41	OK	1.09
1.2 DL + 1.6 WLx + LL	15.83	3.09	7.22	3.09	7.41	45.16	0.95	OK	0.74	0.472	OK	1.25
1.2 DL + 1.6 WLz + LL	14.08	2.75	6.42	2.75	6.59	40.16	0.84	OK	0.74	0.42	OK	1.12
0.9 DL + 1.6 WLx	10.29	2.01	4.69	2.01	4.82	29.35	0.62	OK	0.74	0.307	OK	0.82
0.9 DL + 1.6 WLz	8.53	1.67	3.89	1.67	3.99	24.33	0.51	OK	0.74	0.254	OK	0.68
SUPPORT 4												
1.4 DL	4.52	0.88	2.06	0.88	2.12	12.89	0.27	OK	0.74	0.135	OK	0.36
1.4 DL + 1.7 LL	14.18	2.77	6.47	2.77	6.64	40.45	0.85	OK	0.74	0.423	OK	1.12
1.2 DL + 1.6 WLx + LL	7.50	1.46	3.42	1.46	3.51	21.39	0.45	OK	0.74	0.224	OK	0.59
1.2 DL + 1.6 WLz + LL	14.18	2.77	6.47	2.77	6.64	40.45	0.85	OK	0.74	0.423	OK	1.12
0.9 DL + 1.6 WLx	3.96	0.77	1.81	0.77	1.85	11.30	0.24	OK	0.74	0.118	OK	0.31
0.9 DL + 1.6 WLz	8.52	1.66	3.89	1.66	3.99	24.30	0.51	OK	0.74	0.254	OK	0.68

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Load Case	kn (Ø=0.75)	p =	A_{st}
SUPPORT 1			
1.4 DL	0.45	0.001	169
1.4 DL + 1.7 LL	0.69	0.002	338
1.2 DL + 1.6 WLx + LL	0.67	0.002	338
1.2 DL + 1.6 WLz + LL	0.7	0.002	338
0.9 DL + 1.6 WLx	0.49	0.001	169
0.9 DL + 1.6 WLz	0.58	0.001	169
SUPPORT 2			
1.4 DL	0.45	0.001	169
1.4 DL + 1.7 LL	0.63	0.002	338
1.2 DL + 1.6 WLx + LL	1.21	0.003	507
1.2 DL + 1.6 WLz + LL	0.73	0.002	338
0.9 DL + 1.6 WLx	1.07	0.003	507
0.9 DL + 1.6 WLz	0.59	0.001	169
SUPPORT 3			
1.4 DL	0.55	0.001	169
1.4 DL + 1.7 LL	1.45	0.004	676
1.2 DL + 1.6 WLx + LL	1.67	0.004	676
1.2 DL + 1.6 WLz + LL	1.49	0.004	676
0.9 DL + 1.6 WLx	1.09	0.003	507
0.9 DL + 1.6 WLz	0.9	0.002	338
SUPPORT 4			
1.4 DL	0.48	0.001	169
1.4 DL + 1.7 LL	1.5	0.004	676
1.2 DL + 1.6 WLx + LL	0.79	0.002	338
1.2 DL + 1.6 WLz + LL	1.5	0.004	676
0.9 DL + 1.6 WLx	0.42	0.001	169
0.9 DL + 1.6 WLz	0.9	0.002	338

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	2.23	0.91	1.21	0.003	676.00	304.20	16ϕT	297.43	300	200
At top	0.00	0.00	0.00	0.0018	282.60	282.60	12ϕT	400.20	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

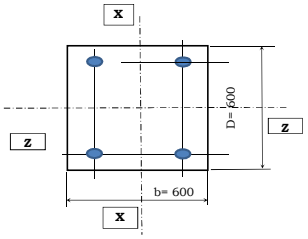
Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

DESIGN OF COLUMN MKD C1, C3 & C8

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1900

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsuported length of column about X (l_{ux}) = 1900.00
Unsupported length of column about Z (l_{uz}) = 1900.00

For Columns not braced against sidesway,
Slenderness ratio, $K\,l_{ux} / r_x$ = 21.95 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 21.95 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 173
Radius of gyration r_z= 173

γ_y =	0.71
γ_z =	0.71

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
SUPPORT 1							
1.4 DL	0.00	3.92	0.14	0.0	6.0	0.4	0.00
1.4 DL + 1.7 LL	0	10.55	0.31	0.0	16.2	0.9	0.01
1.2 DL + 1.6 WLx + LL	-1.92	-2.02	0.22	5.6	-3.1	0.6	0.00
1.2 DL + 1.6 WLz + LL	0	2.94	-1.54	0.0	4.5	4.5	0.00
0.9 DL + 1.6 WLx	-1.92	-6.76	0.09	5.6	-10.4	0.3	-0.01
0.9 DL + 1.6 WLz	0	-1.8	-1.67	0.0	-2.8	4.9	0.00
SUPPORT 2							
1.4 DL	0	4.76	0	0.0	7.3	0.0	0.01
1.4 DL + 1.7 LL	0	9.69	-0.17	0.0	14.9	0.5	0.01
1.2 DL + 1.6 WLx + LL	-1.92	16.26	-0.1	5.6	25.0	0.3	0.02
1.2 DL + 1.6 WLz + LL	0	2.98	-1.7	0.0	4.6	5.0	0.00
0.9 DL + 1.6 WLx	-1.92	12.34	0	5.6	19.0	0.0	0.02
0.9 DL + 1.6 WLz	0	-0.94	-1.6	0.0	-1.4	4.7	0.00
SUPPORT 3							
1.4 DL	-0.28	5.18	-0.14	0.8	8.0	0.4	0.01
1.4 DL + 1.7 LL	-1.47	25.75	-0.31	4.3	39.6	0.9	0.03
1.2 DL + 1.6 WLx + LL	-2.7	25.5	-0.22	7.9	39.2	0.6	0.03
1.2 DL + 1.6 WLz + LL	-0.94	20.54	-1.82	2.7	31.6	5.3	0.03
0.9 DL + 1.6 WLx	-1.94	12.29	-0.09	5.7	18.9	0.3	0.02
0.9 DL + 1.6 WLz	-0.18	7.33	-1.69	0.5	11.3	4.9	0.01
SUPPORT 4							
1.4 DL	0.28	3.92	0	0.8	6.0	0.0	0.00
1.4 DL + 1.7 LL	1.3	29.25	0.17	3.8	45.0	0.5	0.04
1.2 DL + 1.6 WLx + LL	-0.92	9.3	0.1	2.7	14.3	0.3	0.01
1.2 DL + 1.6 WLz + LL	0.84	22.58	-1.66	2.5	34.7	4.9	0.03
0.9 DL + 1.6 WLx	-1.58	-6.44	0	4.6	-9.9	0.0	-0.01
0.9 DL + 1.6 WLz	0.18	6.84	-1.76	0.5	10.5	5.1	0.01

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. = 3928.57 Sq mm
2-Nos	25 mm Dia	982.14 Sq mm	(At middle)	
2-Nos	25 mm Dia	982.14 Sq mm	(At middle)	Total A_{sc} = 1.1 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'c (A_g - A_{sc}) + f_y A_{sc} = 1224.3125 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cbD^2$						
SUPPORT 1								
1.4 DL	0.06	0.06	45.4	45.36	0.55	1.16	0.00	O.K
1.4 DL + 1.7 LL	0.06	0.06	45.4	45.36	0.55	1.16	0.01	O.K
1.2 DL + 1.6 WLx + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.10	O.K
1.2 DL + 1.6 WLz + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.07	O.K
0.9 DL + 1.6 WLx	0.06	0.06	45.4	45.36	0.55	1.16	0.09	O.K
0.9 DL + 1.6 WLz	0.06	0.06	45.4	45.36	0.55	1.16	0.08	O.K
SUPPORT 2								
1.4 DL	0.06	0.06	45.4	45.36	0.55	1.16	0.00	O.K
1.4 DL + 1.7 LL	0.06	0.06	45.4	45.36	0.55	1.16	0.01	O.K
1.2 DL + 1.6 WLx + LL	0.05	0.05	37.8	37.80	0.55	1.16	0.11	O.K
1.2 DL + 1.6 WLz + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.08	O.K
0.9 DL + 1.6 WLx	0.05	0.05	37.8	37.80	0.55	1.16	0.11	O.K
0.9 DL + 1.6 WLz	0.06	0.06	45.4	45.36	0.55	1.16	0.07	O.K
SUPPORT 3								
1.4 DL	0.06	0.06	45.4	45.36	0.55	1.16	0.01	O.K
1.4 DL + 1.7 LL	0.06	0.06	45.4	45.36	0.55	1.16	0.08	O.K
1.2 DL + 1.6 WLx + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.14	O.K
1.2 DL + 1.6 WLz + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.12	O.K
0.9 DL + 1.6 WLx	0.05	0.05	37.8	37.80	0.55	1.16	0.11	O.K
0.9 DL + 1.6 WLz	0.06	0.06	45.4	45.36	0.55	1.16	0.08	O.K
SUPPORT 4								
1.4 DL	0.06	0.06	45.4	45.36	0.55	1.16	0.01	O.K
1.4 DL + 1.7 LL	0.06	0.06	45.4	45.36	0.55	1.16	0.06	O.K
1.2 DL + 1.6 WLx + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.04	O.K
1.2 DL + 1.6 WLz + LL	0.06	0.06	45.4	45.36	0.55	1.16	0.11	O.K
0.9 DL + 1.6 WLx	0.06	0.06	45.4	45.36	0.55	1.16	0.07	O.K
0.9 DL + 1.6 WLz	0.06	0.06	45.4	45.36	0.55	1.16	0.09	O.K

Links, **10mm dia @ 300 C/C**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

4 3340 TANK SUPPORTING STRUCTURE

4.1 Design of Foundation marked F2

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2194)

FOUNDATION SIZE = 2.50m X 5.50m

DEPTH OF THE FOUNDATION = 2.00m BELOW FGL

THE WT OF SOIL +SELF WT = 2.5X5.5X2X2 = 55 T

WORKING STRESS DESIGN	NODE NO.						Load at bottom of foundation				
	4			3							
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx	Mx	Hz	Mz
DL+LL	-0.35	14.63	-0.08	-0.15	14.12	0.02	83.76	-0.51	1.27	-0.06	1.27
DL+0.75*LL+0.75*WLx	-3.32	18.52	-0.05	-2.89	17.48	0.05	91.00	-6.21	2.28	0.00	15.54
DL+0.75*LL+0.75*WLz	-0.46	20.71	-2.75	-0.03	3.86	-2.65	79.57	-0.49	50.58	-5.40	1.22
0.6*DL+WLx	-4.16	11.21	0.00	-3.74	10.15	0.04	76.36	-7.90	2.41	0.04	19.76
0.6*DL+WLz	-0.34	14.14	-3.60	0.07	-8.01	-3.56	61.13	-0.27	66.62	-7.16	0.67

Check for overturning and sliding:

Factor of safety against both Overturning & Sliding is 1.5.

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	83.76	-0.51	1.27	-0.06	1.27	82.44	180.81	65.95	567.84
DL+0.75*LL+0.75*WLx	91.00	-6.21	15.54	0.00	2.28	7.32	109.72	5.86	29118.64
DL+0.75*LL+0.75*WLz	79.57	-0.49	1.22	-5.40	50.58	81.24	4.33	64.99	5.90
0.6*DL+WLx	76.36	-7.90	19.76	0.04	2.41	4.83	87.10	3.87	848.44
0.6*DL+WLz	61.13	-0.27	0.67	-7.16	66.62	113.62	2.52	90.90	3.41

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 16T/m2 IN LC 3 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	0	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

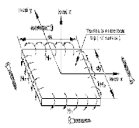
Foundation Thickness =	350	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	4.774	T-M	Corresponding Mxy	0.44	T-M	*(Plate No. - 323 , L/C No. - 105)*
Min. Mx	-9.394	T-M	Corresponding Mxy	0.22	T-M	*(Plate No. - 331 , L/C No. - 103)*
Max. My	7.844	T-M	Corresponding Mxy	-1.20	T-M	*(Plate No. - 31 , L/C No. - 103)*
Min. My	-6.808	T-M	Corresponding Mxy	-0.55	T-M	*(Plate No. - 196 , L/C No. - 105)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	4.774 + 0.44 =	5.214 T-M
My1 =	My + abs(Mxy) =	7.844 + 1.197 =	9.041 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	4.774 + 0.025 =	4.799 T-M
My2 =	My+abs(Mxy^2/Mx) =	7.844 + 0.3 =	8.144 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 5.214 T-m
Myd = 9.041 T-m

ALONG X - DIRECTION :

Mu = 5.214 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5.214X10^7/(1000X251^2) = 0.828 N/mm^2 < deff. = 251 mm = 6.9 N/mm^2
Pt = 0.0022 Reqd. Ast = 0.0022X1000X251/100 = 5.52 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 9.041 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 9.041X10^7/(1000X267^2) = 1.268 N/mm^2 < deff. = 267 mm = 6.9 N/mm^2
Pt = 0.0034 Reqd. Ast = 0.0034X1000X267/100 = 9.08 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-9.394 - 0.224 =	-9.618 T-M
My1 =	My - abs(Mxy) =	-6.808 - 0.552 =	-7.36 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-9.394 - 0.007 =	-9.401 T-M
My2 =	My-abs(Mxy^2/Mx) =	-6.808 - 0.032 =	-6.776 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -9.618 T-M
Myd = -7.36 T-M

ALONG X - DIRECTION :

Mu = 9.618 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 9.618X10^7/(1000X251^2) = 1.527 N/mm^2 < deff. = 251 mm = 6.9 N/mm^2
Pt = 0.0042 Reqd. Ast = 0.0042X1000X251/100 = 10.54 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 7.36 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 7.36X10^7/(1000X267^2) = 1.032 N/mm^2 < deff. = 267 mm = 6.9 N/mm^2
Pt = 0.0028 Reqd. Ast = 0.0028X1000X267/100 = 7.48 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.467 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 333 , L/C No. - 103)*
Vu = 11.722 T

Provided Pt = 0.0053
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0053X11.722X10^4X251/9.618/10^7)X1000X251/10^4
= 24.451 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0053)X1000X251/10^4
= 26.02 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X251/10^4
= 43.063 T
HENCE, Vc = 24.451 T
Φt*Vc = 0.75X24.451 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 18.338 T **> 11.722T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.261 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 186 , L/C No. - 102)*
Vu = 6.969 T

Pt provided = 0.005
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.005X6.969X10^4X267/7.36/10^7)X1000X267/10^4
= 25.847 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.005)X1000X267/10^4
= 27.543 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X267/10^4
= 45.808 T
HENCE, Vc = 25.847 T
Φt*Vc = 0.75X25.847 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 19.385 T **> 6.969T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F2

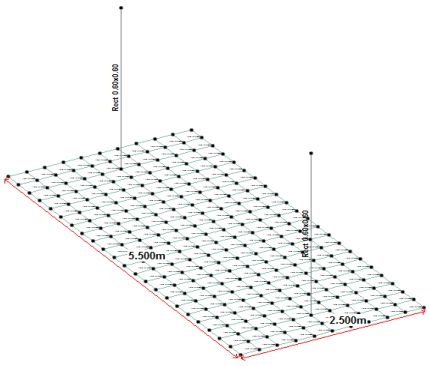
Size of Column = 600 X 600 mm^2

Reqired area of Reinforcement = 3600 mm^2

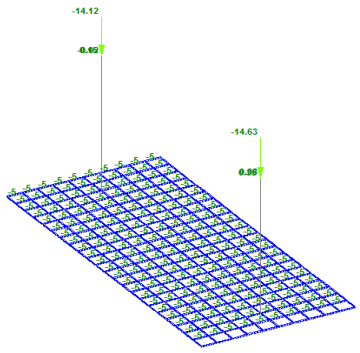
Reinforcement Provided = 8 no. 25 dia bars

Area of Reinforcement Provided = 3927 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 2 T/m^2
Soil Weight over the foundation = 1.65x 1.8 T/m2 = 2.97 T/m^2
SOIL+SURCHARGE LOAD = 4.97 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
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Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI		

4.2 Design of Foundation marked F3, F6A & F8

(Refer Calculation No. MD-216-3340-EG-ST-CAL-2194)

	173 (T)			176 (T)			217 (T)			218 (T)		
Load Detail	FX	FY	FZ	FX	FY	FZ	FX	FY	FZ	FX	FY	FZ
DEAD LOAD	0.0	2.8	-0.2	0.0	2.8	0.2	0.0	5.4	0.0	0.0	2.9	0.0
LIVE LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
EQUIPMENT DEAD LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
EQUIPMENT LIVE LOAD	0.0	0.6	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
WIND LOAD_X	-0.1	0.0	0.0	-0.1	0.0	0.0	-0.1	0.0	0.0	-0.1	0.0	0.0
WIND LOAD_Z	0.0	0.1	-0.1	0.0	-0.1	-0.1	0.0	0.0	-0.1	0.0	0.0	-0.1

STRENGTH DESIGN			
Load Combinations	FX (T)	FY (T)	FZ (T)
SUPPORT 1			
1.4 DL	0.0	3.9	-0.2
1.4 DL + 1.7 LL	0.0	4.9	-0.3
1.2 DL + 1.6 WLx + LL	-0.2	3.9	-0.2
1.2 DL + 1.6 WLz + LL	0.0	4.1	-0.4
0.9 DL + 1.6 WLx	-0.2	2.5	-0.2
0.9 DL + 1.6 WLz	0.0	2.6	-0.3
SUPPORT 2			
1.4 DL	0.0	3.9	0.2
1.4 DL + 1.7 LL	0.0	4.9	0.3
1.2 DL + 1.6 WLx + LL	-0.2	4.5	0.2
1.2 DL + 1.6 WLz + LL	0.0	4.3	0.1
0.9 DL + 1.6 WLx	-0.2	2.5	0.2
0.9 DL + 1.6 WLz	0.0	2.4	0.0
SUPPORT 3			
1.4 DL	0.0	7.6	0.0
1.4 DL + 1.7 LL	0.0	7.6	0.0
1.2 DL + 1.6 WLx + LL	-0.2	7.5	0.0
1.2 DL + 1.6 WLz + LL	0.0	7.6	-0.1
0.9 DL + 1.6 WLx	-0.2	4.8	0.0
0.9 DL + 1.6 WLz	0.0	4.9	-0.1
SUPPORT 4			
1.4 DL	0.0	4.0	0.0
1.4 DL + 1.7 LL	0.0	4.0	0.0
1.2 DL + 1.6 WLx + LL	-0.2	4.0	0.0
1.2 DL + 1.6 WLz + LL	0.0	4.0	-0.1
0.9 DL + 1.6 WLx	-0.2	2.5	0.0
0.9 DL + 1.6 WLz	0.0	2.6	-0.1

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisic	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
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DESIGN OF FOUNDATION

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.20m	1.20m
Thickness of raft :	0.25 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

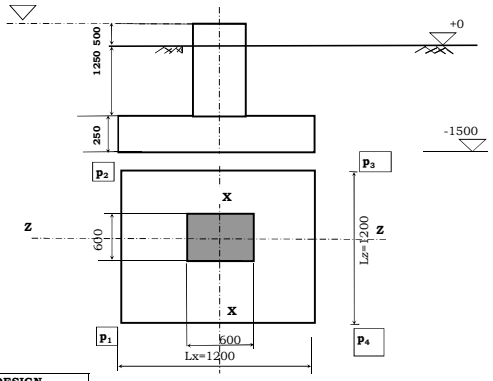
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	WL

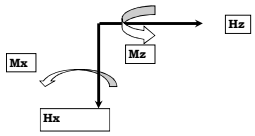
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.2*1.2*0.25*2.5)+(0.6*0.6*1.75*2.5) = 2.48T

Weight of soil = ((1.2*1.2)-(0.6*0.6))*1.25*1.8 = 2.43T

U/S Base Pl Total = 4.91T





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 169mm = 0.169m

Effective depth of slab along Z direction = 157mm = 0.157m

Clear projection along Z direction = 0.3m

Clear projection along X direction = 0.3m

WORKING STRESS DESIGN

Load Case	Load at U/S Of Base Plt					Load at bottom of foundation				
SUPPORT 1	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	3.35	-0.18	0.0	0.00	8.26	0.00	0.36	-0.18	0.00
DL+0.75*LL+0.75*WLx	-0.09	3.21	-0.18	0.0	0.00	8.12	-0.09	0.36	-0.18	0.18
DL+0.75*LL+0.75*WLz	0.00	3.27	-0.25	0.0	0.00	8.18	0.00	0.50	-0.25	0.00
0.6*DL+WLx	-0.13	1.67	-0.10	0.0	0.00	6.58	-0.13	0.20	-0.10	0.26
0.6*DL+WLz	0.00	1.76	-0.19	0.0	0.00	6.67	0.00	0.38	-0.19	0.00
SUPPORT 2										
DL+LL	0.00	3.34	0.18	0.0	0.00	8.25	0.00	0.36	0.18	0.00
DL+0.75*LL+0.75*WLx	-0.09	3.20	0.18	0.0	0.00	8.11	-0.09	0.36	0.18	0.18
DL+0.75*LL+0.75*WLz	0.00	3.13	0.11	0.0	0.00	8.04	0.00	0.22	0.11	0.00
0.6*DL+WLx	-0.13	1.67	0.10	0.0	0.00	6.58	-0.13	0.20	0.10	0.26
0.6*DL+WLz	0.00	1.58	0.01	0.0	0.00	6.49	0.00	0.02	0.01	0.00
SUPPORT 3										
DL+LL	-0.03	5.41	0.00	0.0	0.00	10.32	-0.03	0.00	0.00	0.06
DL+0.75*LL+0.75*WLx	-0.11	5.39	0.00	0.0	0.00	10.30	-0.11	0.00	0.00	0.22
DL+0.75*LL+0.75*WLz	-0.03	5.41	-0.05	0.0	0.00	10.32	-0.03	0.10	-0.05	0.06
0.6*DL+WLx	-0.12	3.21	0.00	0.0	0.00	8.12	-0.12	0.00	0.00	0.24
0.6*DL+WLz	-0.02	3.24	-0.06	0.0	0.00	8.15	-0.02	0.12	-0.06	0.04
SUPPORT 4										
DL+LL	-0.04	2.88	0.00	0.0	0.00	7.79	-0.04	0.00	0.00	0.08
DL+0.75*LL+0.75*WLx	-0.11	2.85	0.00	0.0	0.00	7.76	-0.11	0.00	0.00	0.22
DL+0.75*LL+0.75*WLz	-0.04	2.88	-0.05	0.0	0.00	7.79	-0.04	0.10	-0.05	0.08
0.6*DL+WLx	-0.12	1.69	0.00	0.0	0.00	6.60	-0.12	0.00	0.00	0.24
0.6*DL+WLz	-0.02	1.72	-0.06	0.0	0.00	6.63	-0.02	0.12	-0.06	0.04

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
SUPPORT 1									
DL+LL	8.26	0.00	0.00	-0.18	0.36	NA	13.77	NA	18.36
DL+0.75*LL+0.75*WLx	8.12	-0.09	0.18	-0.18	0.36	27.07	13.53	36.09	18.04
DL+0.75*LL+0.75*WLz	8.18	0.00	0.00	-0.25	0.50	NA	9.82	NA	13.09
0.6*DL+WLx	6.58	-0.13	0.26	-0.10	0.20	15.18	19.74	20.25	26.32
0.6*DL+WLz	6.67	0.00	0.00	-0.19	0.38	NA	10.53	NA	14.04
SUPPORT 2									
DL+LL	8.25	0.00	0.00	0.18	0.36	NA	13.75	NA	18.33
DL+0.75*LL+0.75*WLx	8.11	-0.09	0.18	0.18	0.36	27.03	13.52	36.04	18.02
DL+0.75*LL+0.75*WLz	8.04	0.00	0.00	0.11	0.22	NA	21.93	NA	29.24
0.6*DL+WLx	6.58	-0.13	0.26	0.10	0.20	15.18	19.74	20.25	26.32
0.6*DL+WLz	6.49	0.00	0.00	0.01	0.02	NA	194.70	NA	259.60
SUPPORT 3									
DL+LL	10.32	-0.03	0.06	0.00	0.00	103.20	NA	137.60	NA
DL+0.75*LL+0.75*WLx	10.30	-0.11	0.22	0.00	0.00	28.09	NA	37.45	NA
DL+0.75*LL+0.75*WLz	10.32	-0.03	0.06	-0.05	0.10	103.20	61.92	137.60	82.56
0.6*DL+WLx	8.12	-0.12	0.24	0.00	0.00	20.30	NA	27.07	NA
0.6*DL+WLz	8.15	-0.02	0.04	-0.06	0.12	122.25	40.75	163.00	54.33
SUPPORT 4									
DL+LL	7.79	-0.04	0.08	0.00	0.00	58.43	NA	77.90	NA
DL+0.75*LL+0.75*WLx	7.76	-0.11	0.22	0.00	0.00	21.16	NA	28.22	NA
DL+0.75*LL+0.75*WLz	7.79	-0.04	0.08	-0.05	0.10	58.43	46.74	77.90	62.32
0.6*DL+WLx	6.60	-0.12	0.24	0.00	0.00	16.50	NA	22.00	NA
0.6*DL+WLz	6.63	-0.02	0.04	-0.06	0.12	99.45	33.15	132.60	44.20

Factor of safety against both Overturning & Sliding is 1.5. Hence it is OK.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023			Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC			Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI		

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
SUPPORT 1	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4 DL	0.00	3.89	-0.24	0.0	0.00	14.492	0.48	0.00
1.4 DL + 1.7 LL	0.00	4.86	-0.27	0.0	0.00	15.461	0.54	0.00
1.2 DL + 1.6 WLx + LL	-0.21	3.92	-0.22	0.0	0.00	14.522	0.45	0.42
1.2 DL + 1.6 WLz + LL	0.00	4.05	-0.37	0.0	0.00	14.650	0.74	0.00
0.9 DL + 1.6 WLx	-0.21	2.52	-0.15	0.0	0.00	13.118	0.31	0.42
0.9 DL + 1.6 WLz	0.00	2.65	-0.30	0.0	0.00	13.246	0.59	0.00
SUPPORT 2								
1.4 DL	0.00	3.91	0.24	0.0	0.00	14.506	0.48	0.00
1.4 DL + 1.7 LL	0.00	4.86	0.26	0.0	0.00	15.458	0.51	0.00
1.2 DL + 1.6 WLx + LL	-0.21	4.45	0.25	0.0	0.00	15.050	0.50	0.42
1.2 DL + 1.6 WLz + LL	0.00	4.32	0.10	0.0	0.00	14.922	0.21	0.00
0.9 DL + 1.6 WLx	-0.21	2.50	0.15	0.0	0.00	13.095	0.31	0.42
0.9 DL + 1.6 WLz	0.00	2.37	0.01	0.0	0.00	12.967	0.02	0.00
SUPPORT 3								
1.4 DL	-0.04	7.57	0.00	0.0	0.00	18.174	0.00	0.08
1.4 DL + 1.7 LL	-0.04	7.57	0.00	0.0	0.00	18.174	0.00	0.08
1.2 DL + 1.6 WLx + LL	-0.20	7.53	0.00	0.0	0.00	18.126	0.00	0.40
1.2 DL + 1.6 WLz + LL	-0.04	7.57	-0.10	0.0	0.00	18.174	0.19	0.08
0.9 DL + 1.6 WLx	-0.19	4.82	0.00	0.0	0.00	15.421	0.00	0.37
0.9 DL + 1.6 WLz	-0.03	4.87	-0.10	0.0	0.00	15.469	0.19	0.05
SUPPORT 4								
1.4 DL	-0.04	4.02	0.00	0.0	0.00	14.618	0.00	0.08
1.4 DL + 1.7 LL	-0.04	4.02	0.00	0.0	0.00	14.618	0.00	0.08
1.2 DL + 1.6 WLx + LL	-0.20	3.97	0.00	0.0	0.00	14.570	0.00	0.40
1.2 DL + 1.6 WLz + LL	-0.04	4.02	-0.10	0.0	0.00	14.618	0.19	0.08
0.9 DL + 1.6 WLx	-0.19	2.54	0.00	0.0	0.00	13.135	0.00	0.37
0.9 DL + 1.6 WLz	-0.03	2.58	-0.10	0.0	0.00	13.183	0.19	0.05

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

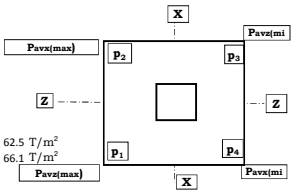
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



SUPPORT 1	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	8.26	0.00	0.36	2.41	0.00	0.34	4.75	4.07	4.75	4.07	2.50	O.K
DL+0.75*LL+0.75*WLx	8.12	0.18	0.36	2.37	0.17	0.34	4.88	4.20	4.54	3.86	2.63	O.K
DL+0.75*LL+0.75*WLz	8.18	0.00	0.50	2.39	0.00	0.47	4.86	3.92	4.86	3.92	2.61	O.K
0.6*DL+WLx	6.58	0.26	0.20	1.92	0.25	0.19	4.36	3.98	3.87	3.49	2.11	O.K
0.6*DL+WLz	6.67	0.00	0.38	1.95	0.00	0.36	4.31	3.59	4.31	3.59	2.06	O.K
SUPPORT 2												
DL+LL	8.25	0.00	0.36	2.41	0.00	0.34	4.75	4.07	4.75	4.07	2.50	O.K
DL+0.75*LL+0.75*WLx	8.11	0.18	0.36	2.37	0.17	0.34	4.88	4.20	4.54	3.86	2.63	O.K
DL+0.75*LL+0.75*WLz	8.04	0.00	0.22	2.35	0.00	0.21	4.56	4.14	4.56	4.14	2.31	O.K
0.6*DL+WLx	6.58	0.26	0.20	1.92	0.25	0.19	4.36	3.98	3.87	3.49	2.11	O.K
0.6*DL+WLz	6.49	0.00	0.02	1.90	0.00	0.02	3.92	3.88	3.92	3.88	1.67	O.K
SUPPORT 3												
DL+LL	10.32	0.06	0.00	3.02	0.06	0.00	5.07	5.07	4.96	4.96	2.82	O.K
DL+0.75*LL+0.75*WLx	10.30	0.22	0.00	3.01	0.21	0.00	5.22	5.22	4.80	4.80	2.97	O.K
DL+0.75*LL+0.75*WLz	10.32	0.06	0.10	3.02	0.06	0.09	5.17	4.98	5.05	4.86	2.92	O.K
0.6*DL+WLx	8.12	0.24	0.00	2.37	0.23	0.00	4.60	4.60	4.15	4.15	2.35	O.K
0.6*DL+WLz	8.15	0.04	0.12	2.38	0.04	0.11	4.53	4.31	4.46	4.23	2.28	O.K
SUPPORT 4												
DL+LL	7.79	0.08	0.00	2.28	0.08	0.00	4.35	4.35	4.20	4.20	2.10	O.K
DL+0.75*LL+0.75*WLx	7.76	0.22	0.00	2.27	0.21	0.00	4.48	4.48	4.06	4.06	2.23	O.K
DL+0.75*LL+0.75*WLz	7.79	0.08	0.10	2.28	0.08	0.09	4.45	4.26	4.30	4.11	2.20	O.K
0.6*DL+WLx	6.60	0.24	0.00	1.93	0.23	0.00	4.16	4.16	3.70	3.70	1.91	O.K
0.6*DL+WLz	6.63	0.04	0.12	1.94	0.04	0.11	4.09	3.86	4.01	3.79	1.84	O.K

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
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Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
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Outotec / L&T Revisic	0	1	2	3
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Calculation of Design Pressure

Load Combination & Calculation of Base pressure

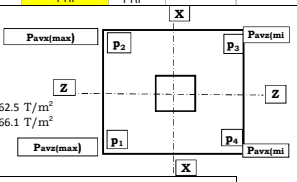
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



SUPPORT 1	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4 DL	14.5	0.00	0.48	4.23	0.00	0.45	6.69	5.78	6.69	5.78	4.44
1.4 DL + 1.7 LL	15.5	0.00	0.54	4.52	0.00	0.52	7.03	4.00	5.03	4.00	4.78
1.2 DL + 1.6 WLx + LL	14.5	0.42	0.45	4.24	0.39	0.42	7.06	4.21	4.27	3.42	4.81
1.2 DL + 1.6 WLz + LL	14.7	0.00	0.74	4.28	0.00	0.70	6.98	3.58	4.98	3.58	4.73
0.9 DL + 1.6 WLx	13.1	0.42	0.31	3.83	0.39	0.29	6.52	3.94	3.73	3.15	4.27
0.9 DL + 1.6 WLz	13.2	0.00	0.59	3.87	0.00	0.56	6.43	3.31	4.43	3.31	4.18
SUPPORT 2											
1.4 DL	14.5	0.00	0.48	4.24	0.00	0.45	6.69	3.79	4.69	3.79	4.44
1.4 DL + 1.7 LL	15.5	0.00	0.51	4.52	0.00	0.48	7.00	4.03	5.00	4.03	4.75
1.2 DL + 1.6 WLx + LL	15.1	0.42	0.50	4.40	0.39	0.47	7.26	4.32	4.47	3.53	5.01
1.2 DL + 1.6 WLz + LL	14.9	0.00	0.21	4.36	0.00	0.20	6.56	4.16	4.56	4.16	4.31
0.9 DL + 1.6 WLx	13.1	0.42	0.31	3.83	0.39	0.29	6.51	3.93	3.72	3.14	4.26
0.9 DL + 1.6 WLz	13.0	0.00	0.02	3.79	0.00	0.02	5.81	3.77	3.81	3.77	3.56
SUPPORT 3											
1.4 DL	18.2	0.08	0.00	5.31	0.08	0.00	7.39	5.39	5.23	5.23	5.14
1.4 DL + 1.7 LL	18.2	0.08	0.00	5.31	0.08	0.00	7.39	5.39	5.23	5.23	5.14
1.2 DL + 1.6 WLx + LL	18.1	0.40	0.00	5.30	0.38	0.00	7.68	5.68	4.91	4.91	5.43
1.2 DL + 1.6 WLz + LL	18.2	0.08	0.19	5.31	0.08	0.18	7.57	5.21	5.41	5.05	5.32
0.9 DL + 1.6 WLx	15.4	0.37	0.00	4.51	0.35	0.00	6.86	4.86	4.15	4.15	4.61
0.9 DL + 1.6 WLz	15.5	0.05	0.19	4.52	0.05	0.18	6.75	4.39	4.65	4.29	4.50
SUPPORT 4											
1.4 DL	14.6	0.08	0.00	4.27	0.08	0.00	6.35	4.35	4.19	4.19	4.10
1.4 DL + 1.7 LL	14.6	0.08	0.00	4.27	0.08	0.00	6.35	4.35	4.19	4.19	4.10
1.2 DL + 1.6 WLx + LL	14.6	0.40	0.00	4.26	0.38	0.00	6.64	4.64	3.87	3.87	4.39
1.2 DL + 1.6 WLz + LL	14.6	0.08	0.19	4.27	0.08	0.18	6.53	4.17	4.37	4.01	4.28
0.9 DL + 1.6 WLx	13.1	0.37	0.00	3.84	0.35	0.00	6.19	4.19	3.48	3.48	3.94
0.9 DL + 1.6 WLz	13.2	0.05	0.19	3.85	0.05	0.18	6.08	3.72	3.98	3.62	3.83

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

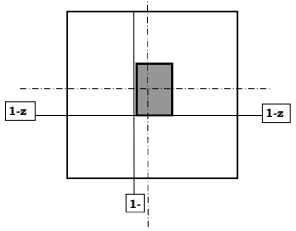
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.6+2*0.5*0.157)+(0.6+2*0.5*0.157))*2

= 3.028 m

Punching Area= ((0.6+2*0.5*0.157)*(0.6+2*0.5*0.157))

= 0.57 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.25*2.5+1.25*1.8)

= 2.88 T/m^2

Area of footing, A_f= 1.44 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 15.49 T

Strength of Concrete in shear Vc1 = 101.67 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 95.48 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 93.75 T **Table 22.6.5.2_ (c)**

Design Vc= 93.75 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ= (15.49*10000)/(3028*157)

= 0.33 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	B.M_{1x} T-m	S.$F_{(oneway)}$ (T)	B.M_{1z}	S.$F_{(Oneway)}$ (T)	S.$F_{(punching)}$	V_u/b_od	Punching check (Remarks)	Allowable shear (0.75*$\sqrt{}$(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
SUPPORT 1												
1.4 DL	4.44	0.87	2.02	0.87	2.08	12.67	0.27	OK	0.74	0.132	OK	0.35
1.4 DL + 1.7 LL	4.78	0.93	2.18	0.93	2.24	13.63	0.29	OK	0.74	0.142	OK	0.38
1.2 DL + 1.6 WLx + LL	4.81	0.94	2.19	0.94	2.25	13.72	0.29	OK	0.74	0.143	OK	0.38
1.2 DL + 1.6 WLz + LL	4.73	0.92	2.16	0.92	2.21	13.49	0.28	OK	0.74	0.141	OK	0.37
0.9 DL + 1.6 WLx	4.27	0.83	1.95	0.83	2.00	12.18	0.26	OK	0.74	0.127	OK	0.34
0.9 DL + 1.6 WLz	4.18	0.82	1.91	0.82	1.96	11.92	0.25	OK	0.74	0.125	OK	0.33
SUPPORT 2												
1.4 DL	4.44	0.87	2.02	0.87	2.08	12.67	0.27	OK	0.74	0.132	OK	0.35
1.4 DL + 1.7 LL	4.75	0.93	2.17	0.93	2.22	13.55	0.29	OK	0.74	0.142	OK	0.38
1.2 DL + 1.6 WLx + LL	5.01	0.98	2.28	0.98	2.34	14.29	0.3	OK	0.74	0.149	OK	0.40
1.2 DL + 1.6 WLz + LL	4.31	0.84	1.97	0.84	2.02	12.29	0.26	OK	0.74	0.128	OK	0.34
0.9 DL + 1.6 WLx	4.26	0.83	1.94	0.83	1.99	12.15	0.26	OK	0.74	0.127	OK	0.34
0.9 DL + 1.6 WLz	3.56	0.70	1.62	0.70	1.67	10.15	0.21	OK	0.74	0.106	OK	0.28
SUPPORT 3												
1.4 DL	5.14	1.00	2.34	1.00	2.41	14.66	0.31	OK	0.74	0.153	OK	0.41
1.4 DL + 1.7 LL	5.14	1.00	2.34	1.00	2.41	14.66	0.31	OK	0.74	0.153	OK	0.41
1.2 DL + 1.6 WLx + LL	5.43	1.06	2.48	1.06	2.54	15.49	0.33	OK	0.74	0.162	OK	0.43
1.2 DL + 1.6 WLz + LL	5.32	1.04	2.43	1.04	2.49	15.18	0.32	OK	0.74	0.159	OK	0.42
0.9 DL + 1.6 WLx	4.61	0.90	2.10	0.90	2.16	13.15	0.28	OK	0.74	0.137	OK	0.37
0.9 DL + 1.6 WLz	4.50	0.88	2.05	0.88	2.11	12.84	0.27	OK	0.74	0.134	OK	0.36
SUPPORT 4												
1.4 DL	4.10	0.80	1.87	0.80	1.92	11.70	0.25	OK	0.74	0.122	OK	0.32
1.4 DL + 1.7 LL	4.10	0.80	1.87	0.80	1.92	11.70	0.25	OK	0.74	0.122	OK	0.32
1.2 DL + 1.6 WLx + LL	4.39	0.86	2.00	0.86	2.05	12.52	0.26	OK	0.74	0.131	OK	0.35
1.2 DL + 1.6 WLz + LL	4.28	0.84	1.95	0.84	2.00	12.21	0.26	OK	0.74	0.128	OK	0.34
0.9 DL + 1.6 WLx	3.94	0.77	1.80	0.77	1.84	11.24	0.24	OK	0.74	0.117	OK	0.31
0.9 DL + 1.6 WLz	3.83	0.75	1.75	0.75	1.79	10.93	0.23	OK	0.74	0.114	OK	0.30

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisio	0	1	2	3
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Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Load Case	kn (Ø=0.75)	p =	A_{st}
SUPPORT 1			
1.4 DL	0.47	0.001	169
1.4 DL + 1.7 LL	0.51	0.001	169
1.2 DL + 1.6 WLx + LL	0.51	0.001	169
1.2 DL + 1.6 WLz + LL	0.5	0.001	169
0.9 DL + 1.6 WLx	0.45	0.001	169
0.9 DL + 1.6 WLz	0.44	0.001	169
SUPPORT 2			
1.4 DL	0.47	0.001	169
1.4 DL + 1.7 LL	0.5	0.001	169
1.2 DL + 1.6 WLx + LL	0.53	0.001	169
1.2 DL + 1.6 WLz + LL	0.46	0.001	169
0.9 DL + 1.6 WLx	0.45	0.001	169
0.9 DL + 1.6 WLz	0.38	0.001	169
SUPPORT 3			
1.4 DL	0.54	0.001	169
1.4 DL + 1.7 LL	0.54	0.001	169
1.2 DL + 1.6 WLx + LL	0.57	0.001	169
1.2 DL + 1.6 WLz + LL	0.56	0.001	169
0.9 DL + 1.6 WLx	0.49	0.001	169
0.9 DL + 1.6 WLz	0.48	0.001	169
SUPPORT 4			
1.4 DL	0.43	0.001	169
1.4 DL + 1.7 LL	0.43	0.001	169
1.2 DL + 1.6 WLx + LL	0.46	0.001	169
1.2 DL + 1.6 WLz + LL	0.45	0.001	169
0.9 DL + 1.6 WLx	0.42	0.001	169
0.9 DL + 1.6 WLz	0.4	0.001	169

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	1.00	0.41	0.54	0.001	169.00	304.20	12ϕT	371.79	300	250
At top	0.00	0.00	0.00	0.0018	282.60	282.60	12ϕT	400.20	300	250

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**250**	c/c	At Top	**12ϕT**	**250**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**250**	c/c	At Top	**12ϕT**	**250**	c/c

Support Reactions for Foundations Marked F6A & F8 are less than that of F3, Hence same design applies for F6A & F8

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
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4.3 Design of Foundation marked F4

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2194)

FOUNDATION SIZE = 2.00m X 3.50m

DEPTH OF THE FOUNDATION = 2.00m BELOW FGL

THE WT OF SOIL +SELF WT = 2.0X3.5X2X2 = 28 T

WORKING STRESS DESIGN	NODE NO.										
	4			**3**			**Load at bottom of foundation**				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx	Mx	Hz	Mz
DL+LL	-0.01	4.96	0.01	-0.11	1.66	0.00	34.63	-0.12	3.60	0.01	0.03
DL+0.75*LL+0.75*WLx	0.00	-2.38	0.01	-2.24	9.00	0.00	34.63	-2.24	16.98	0.01	0.02
DL+0.75*LL+0.75*WLz	-0.01	5.23	-0.10	-0.13	1.39	-0.12	34.63	-0.14	4.20	-0.22	0.55
0.6*DL+WLx	0.00	-6.80	0.01	-2.90	10.78	0.00	31.98	-2.90	24.82	0.01	0.02
0.6*DL+WLz	-0.01	3.34	-0.13	-0.09	0.63	-0.16	31.98	-0.10	2.96	-0.29	0.73

Check for overturning and sliding:

Factor of safety against both Overturning & Sliding is 1.5.

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	34.63	-0.12	3.60	0.01	0.03	19.22	3808.75	115.42	1385.00
DL+0.75*LL+0.75*WLx	34.63	-2.24	16.98	0.01	0.02	2.55	4914.52	6.19	1787.10
DL+0.75*LL+0.75*WLz	34.63	-0.14	4.20	-0.22	0.55	10.31	173.92	97.36	63.24
0.6*DL+WLx	31.98	-2.90	24.82	0.01	0.02	1.61	5024.64	4.42	1827.14
0.6*DL+WLz	31.98	-0.10	2.96	-0.29	0.73	13.48	120.45	127.14	43.80

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 13T/m2 IN LC 4 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
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Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
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Prepared By	SHD	SHD	SHD	
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Approved	PRI	PRI	PRI	

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

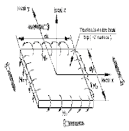
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.662	T-M	Corresponding Mxy	0.13	T-M	*(Plate No. - 323 , L/C No. - 104)*
Min. Mx	-4.052	T-M	Corresponding Mxy	0.12	T-M	*(Plate No. - 331 , L/C No. - 104)*
Max. My	4.16	T-M	Corresponding Mxy	-0.33	T-M	*(Plate No. - 87 , L/C No. - 104)*
Min. My	-5.99	T-M	Corresponding Mxy	-0.28	T-M	*(Plate No. - 87 , L/C No. - 102)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	0.662 + 0.129 =	0.791	T-M
My1 =	My + abs(Mxy) =	4.16 + 0.328 =	4.488	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	0.662 + 0.004 =	0.666	T-M
My2 =	My+abs(Mxy^2/Mx) =	4.16 + 0.163 =	4.323	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 0.791 T-m
Myd = 4.488 T-m

ALONG X - DIRECTION :

Mu = 0.791 T-m
Φ*Kn = Mu/bd^2 deff. = 201 mm
Mu/bd^2 = 0.791X10^7/(1000X201^2) = 0.196 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X201/100 = 1.01 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 4.488 T-m
Φ*Kn = Mu/bd^2 deff. = 217 mm
Mu/bd^2 = 4.488X10^7/(1000X217^2) = 0.953 N/mm^2 < 6.9 N/mm^2
Pt = 0.0026 Reqd. Ast = 0.0026X1000X217/100 = 5.64 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-4.052 - 0.12 =	-4.172	T-M
My1 =	My - abs(Mxy) =	-5.99 - 0.281 =	-6.271	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-4.052 - 0.002 =	-4.054	T-M
My2 =	My-abs(Mxy^2/Mx) =	-5.99 - 0.019 =	-5.971	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -4.172 T-M
Myd = -6.271 T-M

ALONG X - DIRECTION :

Mu = 4.172 T-m
Φ*Kn = Mu/bd^2 deff. = 201 mm
Mu/bd^2 = 4.172X10^7/(1000X201^2) = 1.033 N/mm^2 < 6.9 N/mm^2
Pt = 0.0028 Reqd. Ast = 0.0028X1000X201/100 = 5.63 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 6.271 T-m
Φ*Kn = Mu/bd^2 deff. = 217 mm
Mu/bd^2 = 6.271X10^7/(1000X217^2) = 1.332 N/mm^2 < 6.9 N/mm^2
Pt = 0.0036 Reqd. Ast = 0.0036X1000X217/100 = 7.81 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
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Prepared By	SHD	SHD	SHD	
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SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.164 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 345 , L/C No. - 104)*
Vu = 3.296 T

Provided Pt = 0.005
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.005X3.296X10^4X201/4.172/10^7)X1000X201/10^4
= 19.297 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.005)X1000X201/10^4
= 20.735 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X201/10^4
= 34.485 T
HENCE, Vc = 19.297 T
Φt*Vc = 0.75X19.297 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 14.473 T **> 3.296T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.279 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 76 , L/C No. - 102)*
Vu = 6.054 T

Pt provided = 0.0046
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0046X6.054X10^4X217/6.271/10^7)X1000X217/10^4
= 20.896 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0046)X1000X217/10^4
= 22.238 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X217/10^4
= 37.23 T
HENCE, Vc = 20.896 T
Φt*Vc = 0.75X20.896 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.672 T **> 6.054T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F4

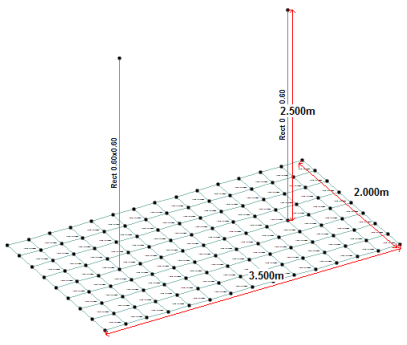
Size of Column = 600 X 600 mm^2

Reqired area of Reinforcement = 3600 mm^2

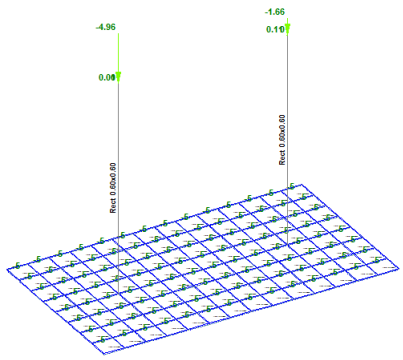
Reinforcement Provided = 8 no. 25 dia bars

Area of Reinforcement Provided = 3927 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 2 T/m^2

Soil Weight over the foundation = 1.7x 1.8 T/m2 = 3.06 T/m^2

SOIL+SURCHARGE LOAD = 5.06 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisic	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI		

5 3340 CONVEYOR CNV-3600-002 SUPPORTING STRUCTURE

5.1 Design of Foundation marked F5

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2194)

FOUNDATION SIZE = 5.875m X 7.30m

DEPTH OF THE FOUNDATION = 1.50m BELOW FGL

THE WT OF SOIL +SELF WT = 5.875X7.3X1.5X2 = 128.6625 T

WORKING STRESS DESIGN	3			415			4			417		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	-0.66	32.40	0.13	-0.14	51.24	-0.09	-0.90	29.18	-0.36	-0.40	38.56	-0.55
DL+0.75*LL+0.75*WLx	-7.04	21.81	0.45	-2.54	50.48	0.23	-7.30	57.98	-1.25	-2.81	49.59	-1.50
DL+0.75*LL+0.75*WLz	-1.46	19.58	-3.71	-0.65	77.96	-4.09	-1.70	24.72	-3.43	-0.93	57.97	-3.78
0.6*DL+WLx	-7.60	-20.12	0.45	-2.95	2.05	0.42	-7.74	32.16	-0.80	-3.09	18.91	-0.92
0.6*DL+WLz	-0.16	-23.09	-5.10	-0.42	38.70	-5.34	-0.27	-12.19	-3.70	-0.58	30.08	-3.95

SLIDING AND OVERTURNING CHECK Factor of safety against both Overturning & Sliding is 1.5.

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	Mx (T-m)	Hz (T)	Mz (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	276.38	-2.1	51.2	-0.88	50.31	19.50	16.14	52.67	125.48
DL+0.75*LL+0.75*WLx	304.86	-19.7	39.6	-2.08	141.72	7.64	6.32	6.19	58.61
DL+0.75*LL+0.75*WLz	305.24	-4.7	190.4	-15.02	52.55	20.62	17.06	25.77	8.13
0.6*DL+WLx	158.00	-21.4	17.3	-0.85	243.25	2.31	1.91	2.96	74.53
0.6*DL+WLz	158.50	-1.4	218.3	-18.10	9.48	59.36	49.12	44.63	3.50

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 26T/m2 IN LC 3 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	29.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

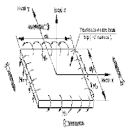
Foundation Thickness =	650	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	23.576	T-M	Corresponding Mxy	1.22	T-M	*(Plate No. - 814 , L/C No. - 103)*
Min. Mx	-13.547	T-M	Corresponding Mxy	0.65	T-M	*(Plate No. - 363 , L/C No. - 102)*
Max. My	23.581	T-M	Corresponding Mxy	-3.28	T-M	*(Plate No. - 750 , L/C No. - 103)*
Min. My	-12.437	T-M	Corresponding Mxy	0.35	T-M	*(Plate No. - 445 , L/C No. - 102)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	23.576 + 1.22 =	24.796	T-M
My1 =	My + abs(Mxy) =	23.581 + 3.278 =	26.859	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	23.576 + 0.063 =	23.639	T-M
My2 =	My+abs(Mxy^2/Mx) =	23.581 + 0.456 =	24.037	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 24.796 T-m
Myd = 26.859 T-m

ALONG X - DIRECTION :

Mu = 24.796 T-m
Φ*Kn = Mu/bd^2 deff. = 545 mm
Mu/bd^2 = 24.796X10^7/(1000X545^2) = 0.835 N/mm^2 < 6.9 N/mm^2
Pt = 0.0022 Reqd. Ast = 0.0022X1000X545/100 = 11.99 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 26.859 T-m
Φ*Kn = Mu/bd^2 deff. = 565 mm
Mu/bd^2 = 26.859X10^7/(1000X565^2) = 0.841 N/mm^2 < 6.9 N/mm^2
Pt = 0.0023 Reqd. Ast = 0.0023X1000X565/100 = 13 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-13.547 - 0.645 =	-14.192	T-M
My1 =	My - abs(Mxy) =	-12.437 - 0.346 =	-12.783	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-13.547 - 0.033 =	-13.58	T-M
My2 =	My-abs(Mxy^2/Mx) =	-12.437 - 0.009 =	-12.428	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -14.192 T-M
Myd = -12.783 T-M

ALONG X - DIRECTION :

Mu = 14.192 T-m
Φ*Kn = Mu/bd^2 deff. = 545 mm
Mu/bd^2 = 14.192X10^7/(1000X545^2) = 0.478 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X545/100 = 7.09 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 12.783 T-m
Φ*Kn = Mu/bd^2 deff. = 565 mm
Mu/bd^2 = 12.783X10^7/(1000X565^2) = 0.4 N/mm^2 < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X565/100 = 6.22 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.3 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 595 , L/C No. - 103)*
Vu = 16.35 T

Provided Pt = 0.0029
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0029X16.35X10^4X545/14.192/10^7)X1000X545/10^4
= 53.275 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0029)X1000X545/10^4
= 54.275 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X545/10^4
= 93.504 T
HENCE, Vc = 53.275 T
Φt*Vc = 0.75X53.275 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 39.956 T **> 16.35T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.7 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 596 , L/C No. - 103)*
Vu = 39.55 T

Pt provided = 0.0028
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X39.55X10^4X565/12.783/10^7)X1000X565/10^4
= 58.183 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X565/10^4
= 56.171 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X565/10^4
= 96.935 T
HENCE, Vc = 56.171 T
Φt*Vc = 0.75X56.171 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 42.128 T **> 39.55T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F5

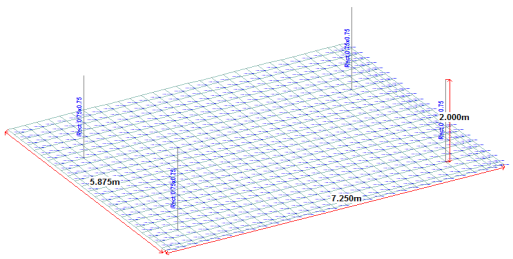
Size of Column = 750 X 750 mm^2

Reqired area of Reinforcement = 5625 mm^2

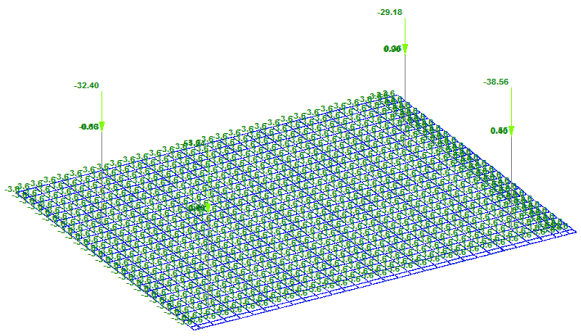
Reinforcement Provided = 12 no. 25 dia bars

Area of Reinforcement Provided = 5890 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation =		2 T/m^2
Soil Weight over the foundation =	0.85 x 1.8 T/m2 =	1.53 T/m^2
SOIL+SURCHARGE LOAD =		3.53 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

6 3340 CONVEYOR CNV-3600-002 TRESTLES & VGTU

6.1 Design of Foundation marked F6

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2193)

FOUNDATION SIZE = 5.000m X 5.00m

DEPTH OF THE FOUNDATION = 2.50m BELOW FGL

THE WT OF SOIL +SELF WT = 5.0X5.0X2.5X2 = 125 T

WORKING STRESS DESIGN	145			146			509			507		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	1.00	32.00	2.00	1.00	33.0	-3.00	-3.00	30.00	2.00	-3.00	37.0	-3.00
DL+0.75*LL+0.75*WLx	-1.75	20.50	2.00	-1.75	21.0	-2.75	-5.75	34.50	2.00	-5.75	40.0	-2.75
DL+0.75*LL+0.75*WLz	2.00	20.50	-1.75	0.50	37.5	-6.50	-1.25	18.00	0.50	-4.25	40.0	-4.25
0.6*DL+WLx	-2.80	0.40	1.25	-2.80	-0.2	-1.20	-5.20	20.00	1.25	-5.20	20.6	-1.20
0.6*DL+WLz	2.20	0.40	-3.80	0.25	21.8	-6.20	0.80	-2.00	-0.80	-3.20	20.6	-3.20

SLIDING AND OVERTURNING CHECK Factor of safety against both Overturning & Sliding is 1.5.

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	Mx (T-m)	Hz (T)	Mz (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	257.00	-4.0	21.6	-2.00	15.20	42.27	29.75	25.70	51.40
DL+0.75*LL+0.75*WLx	241.00	-15.0	16.2	-1.50	108.00	5.58	37.19	6.43	64.27
DL+0.75*LL+0.75*WLz	241.00	-3.0	111.6	-12.00	8.40	71.73	5.40	32.13	8.03
0.6*DL+WLx	165.80	-16.0	0.3	0.10	125.60	3.30	1480.36	4.15	663.20
0.6*DL+WLz	165.80	0.0	127.2	-14.00	7.34	56.47	3.26	1326.40	4.74

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 26T/m2 IN LC 2 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

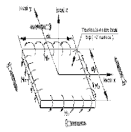
Foundation Thickness =	600	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	7.913	T-M	Corresponding Mxy	-0.62	T-M	*(Plate No. - 393 , L/C No. - 104)*
Min. Mx	-21.847	T-M	Corresponding Mxy	-1.31	T-M	*(Plate No. - 505 , L/C No. - 103)*
Max. My	11.04	T-M	Corresponding Mxy	-1.37	T-M	*(Plate No. - 130 , L/C No. - 105)*
Min. My	-22.001	T-M	Corresponding Mxy	-0.83	T-M	*(Plate No. - 537 , L/C No. - 102)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	7.913 + 0.615 =	8.528	T-M
My1 =	My + abs(Mxy) =	11.04 + 1.365 =	12.405	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	7.913 + 0.034 =	7.947	T-M
My2 =	My+abs(Mxy^2/Mx) =	11.04 + 0.235 =	11.275	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 8.528 T-m
Myd = 12.405 T-m

ALONG X - DIRECTION :

Mu = 8.528 T-m
Φ*Kn = Mu/bd^2 deff. = 501 mm
Mu/bd^2 = 8.528X10^7/(1000X501^2) = 0.34 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X501/100 = 4.51 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 12.405 T-m
Φ*Kn = Mu/bd^2 deff. = 517 mm
Mu/bd^2 = 12.405X10^7/(1000X517^2) = 0.464 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0.0012X1000X517/100 = 6.2 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-21.847 - 1.313 =	-23.16	T-M
My1 =	My - abs(Mxy) =	-22.001 - 0.832 =	-22.833	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-21.847 - 0.078 =	-21.925	T-M
My2 =	My-abs(Mxy^2/Mx) =	-22.001 - 0.032 =	-21.969	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -23.16 T-M
Myd = -22.833 T-M

ALONG X - DIRECTION :

Mu = 23.16 T-m
Φ*Kn = Mu/bd^2 deff. = 501 mm
Mu/bd^2 = 23.16X10^7/(1000X501^2) = 0.923 N/mm^2 < 6.9 N/mm^2
Pt = 0.0025 Reqd. Ast = 0.0025X1000X501/100 = 12.53 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 22.833 T-m
Φ*Kn = Mu/bd^2 deff. = 517 mm
Mu/bd^2 = 22.833X10^7/(1000X517^2) = 0.854 N/mm^2 < 6.9 N/mm^2
Pt = 0.0023 Reqd. Ast = 0.0023X1000X517/100 = 11.89 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.40 cm2) at Horizontal

Total Provide Ast = 13.40 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.7 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 495 , L/C No. - 103)*

Vu = 35.07 T

Provided Pt = 0.0027

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0027X35.07X10^4X501/23.16/10^7)X1000X501/10^4

= 49.168 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0027)X1000X501/10^4

= 49.723 T

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X501/10^4

= 85.955 T

HENCE, Vc = 49.168 T

Φt*Vc = 0.75X49.168 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 36.876 T **> 35.07T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.7 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 533 , L/C No. - 102)*

Vu = 36.19 T

Pt provided = 0.0026

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X36.19X10^4X517/22.833/10^7)X1000X517/10^4

= 50.81 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X517/10^4

= 51.223 KN

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X517/10^4

= 88.7 T

HENCE, Vc = 50.81 T

Φt*Vc = 0.75X50.81 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 38.108 T **> 36.19T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F6

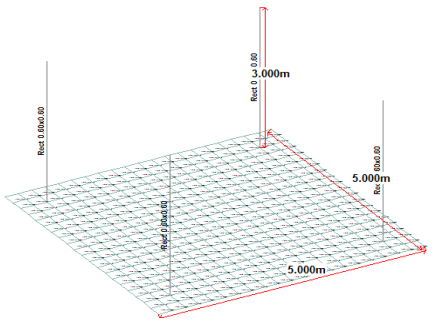
Size of Column = 600 X 600 mm^2

Reqired area of Reinforcement = 3600 mm^2

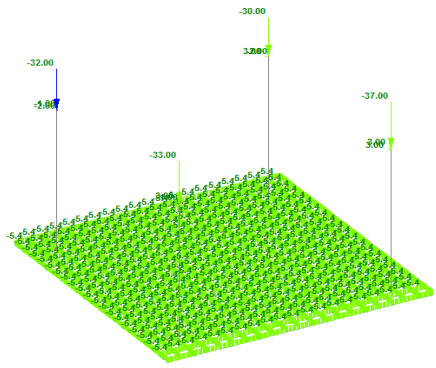
Reinforcement Provided = 8 no. 25 dia bars

Area of Reinforcement Provided = 3927 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation =		2 T/m^2
Soil Weight over the foundation =	1.9 x 1.8 T/m2 =	3.42 T/m^2
SOIL+SURCHARGE LOAD =		5.42 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI	

6.3 Design of Foundation marked F7

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2193)

FOUNDATION SIZE = 1.50m X 2.50m

DEPTH OF THE FOUNDATION = 1.50m BELOW FGL

THE WT OF SOIL +SELF WT = 1.5X2.5X2X2 = 11.25 T

WORKING STRESS DESIGN	NODE NO.										
	4			3			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx	Mx	Hz	Mz
DL+LL	-0.03	11.22	0.17	-0.02	15.86	-1.07	38.33	-0.05	1.68	-0.90	-0.10
DL+0.75*LL+0.75*WLx	-0.07	9.74	0.19	-0.06	13.22	-0.87	34.21	-0.13	1.25	-0.68	-0.26
DL+0.75*LL+0.75*WLz	-0.05	3.77	-1.16	0.01	19.18	-2.22	34.20	-0.04	4.80	-3.38	-0.08
0.6*DL+WLx	-0.06	3.17	0.15	-0.06	3.17	-0.15	17.59	-0.12	0.00	0.00	-0.24
0.6*DL+WLz	-0.04	-4.78	-1.66	0.04	11.13	-1.95	17.60	0.00	4.71	-3.61	0.00

Check for overturning and sliding:

Factor of safety against both Overturning & Sliding is 1.5.

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	38.33	-0.05	1.68	-0.90	-0.10	17.11	479.13	306.64	17.04
DL+0.75*LL+0.75*WLx	34.21	-0.13	1.25	-0.68	-0.26	20.53	164.47	105.26	20.12
DL+0.75*LL+0.75*WLz	34.20	-0.04	4.80	-3.38	-0.08	5.35	534.38	342.00	4.05
0.6*DL+WLx	17.59	-0.12	0.00	0.00	-0.24	NA	91.61	58.63	NA
0.6*DL+WLz	17.60	0.00	4.71	-3.61	0.00	2.80	NA	NA	1.95

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 25T/m2 IN LC 3 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

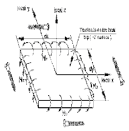
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.739	T-M	Corresponding Mxy	0.06	T-M	*(Plate No. - 331 , L/C No. - 105)*
Min. Mx	-4.517	T-M	Corresponding Mxy	0.08	T-M	*(Plate No. - 322 , L/C No. - 103)*
Max. My	3.316	T-M	Corresponding Mxy	-0.48	T-M	*(Plate No. - 31 , L/C No. - 103)*
Min. My	-1.515	T-M	Corresponding Mxy	0.19	T-M	*(Plate No. - 87 , L/C No. - 105)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	1.739 + 0.058 =	1.797 T-M
My1 =	My + abs(Mxy) =	3.316 + 0.484 =	3.8 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	1.739 + 0.001 =	1.74 T-M
My2 =	My+abs(Mxy^2/Mx) =	3.316 + 0.135 =	3.451 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 1.797 T-m
Myd = 3.8 T-m

ALONG X - DIRECTION :

Mu = 1.797 T-m
Φ*Kn = Mu/bd^2 deff. = 201 mm
Mu/bd^2 = 1.797X10^7/(1000X201^2) = 0.445 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0.0012X1000X201/100 = 2.41 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 3.8 T-m
Φ*Kn = Mu/bd^2 deff. = 217 mm
Mu/bd^2 = 3.8X10^7/(1000X217^2) = 0.807 N/mm^2 < 6.9 N/mm^2
Pt = 0.0022 Reqd. Ast = 0.0022X1000X217/100 = 4.77 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-4.517 - 0.082 =	-4.599 T-M
My1 =	My - abs(Mxy) =	-1.515 - 0.186 =	-1.701 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-4.517 - 0.004 =	-4.521 T-M
My2 =	My-abs(Mxy^2/Mx) =	-1.515 - 0.008 =	-1.507 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -4.599 T-M
Myd = -1.701 T-M

ALONG X - DIRECTION :

Mu = 4.599 T-m
Φ*Kn = Mu/bd^2 deff. = 201 mm
Mu/bd^2 = 4.599X10^7/(1000X201^2) = 1.138 N/mm^2 < 6.9 N/mm^2
Pt = 0.0031 Reqd. Ast = 0.0031X1000X201/100 = 6.23 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 1.701 T-m
Φ*Kn = Mu/bd^2 deff. = 217 mm
Mu/bd^2 = 1.701X10^7/(1000X217^2) = 0.361 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X217/100 = 2.17 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisio	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.099 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 320 , L/C No. - 103)*
Vu = 1.99 T

Provided Pt = 0.005
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.005X1.99X10^4X201/4.599/10^7)X1000X201/10^4
= 19.175 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.005)X1000X201/10^4
= 20.735 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X201/10^4
= 34.485 T
HENCE, Vc = 19.175 T
Φt*Vc = 0.75X19.175 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 14.381 T **> 1.99T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.201 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 73 , L/C No. - 103)*
Vu = 4.362 T

Pt provided = 0.0046
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0046X4.362X10^4X217/1.701/10^7)X1000X217/10^4
= 21.485 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0046)X1000X217/10^4
= 22.238 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X217/10^4
= 37.23 T
HENCE, Vc = 21.485 T
Φt*Vc = 0.75X21.485 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 16.114 T **> 4.362T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F7

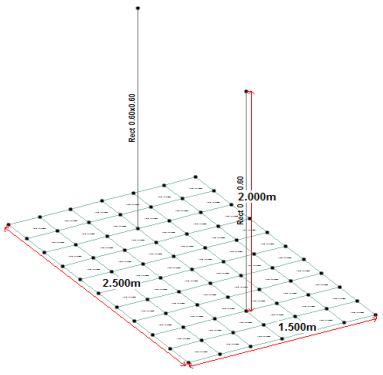
Size of Column = 600 X 600 mm^2

Reqired area of Reinforcement = 3600 mm^2

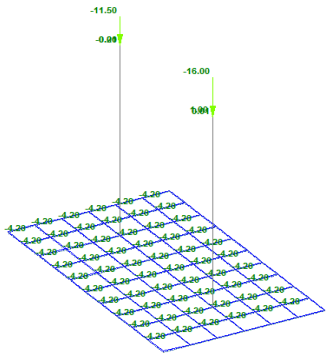
Reinforcement Provided = 8 no. 25 dia bars

Area of Reinforcement Provided = 3927 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 2 T/m^2
Soil Weight over the foundation = 1.2x 1.8 T/m2 = 2.16 T/m^2
SOIL+SURCHARGE LOAD = 4.16 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

6.4 Design of Foundation marked F9

(For Foundation load Data Refer Calculation No. MD-216-3340-EG-ST-CAL-2193)

FOUNDATION SIZE = 3.500m X 3.50m

DEPTH OF THE FOUNDATION = 2.50m BELOW FGL

THE WT OF SOIL +SELF WT = 3.5X3.5X2.5X2 = 61.25 T

WORKING STRESS DESIGN	71			72			73			74		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	2.15	1.80	0.00	2.15	1.8	0.00	2.15	-1.60	0.00	2.15	-1.6	0.00

SLIDING AND OVERTURNING CHECK — Factor of safety against both Overturning & Sliding is 1.5.

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	Mx (T-m)	Hz (T)	Mz (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	61.40	8.6	0.0	0.00	27.48	3.91	N.A.	2.86	N.A.

HENCE THE FOUNDATION IS SAFE IN BOTH SLIDING & OVERTURNING

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m2

MAXIMUM GROSS BEARING PRESSURE OBTAINED 11T/m2 . HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI		

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 105)

REINFORCEMENT CALCULATION OF FOUNDATION:

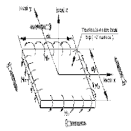
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.958	T-M	Corresponding Mxy	-0.15	T-M	*(Plate No. - 46 , L/C No. - 100)*
Min. Mx	-2.586	T-M	Corresponding Mxy	-0.20	T-M	*(Plate No. - 83 , L/C No. - 100)*
Max. My	0.892	T-M	Corresponding Mxy	0.16	T-M	*(Plate No. - 92 , L/C No. - 100)*
Min. My	-1.632	T-M	Corresponding Mxy	-0.20	T-M	*(Plate No. - 83 , L/C No. - 100)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	1.958 + 0.153 =	2.111	T-M
My1 =	My + abs(Mxy) =	0.892 + 0.159 =	1.051	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	1.958 + 0.026 =	1.984	T-M
My2 =	My+abs(Mxy^2/Mx) =	0.892 + 0.013 =	0.905	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.111 T-m
Myd = 1.051 T-m

ALONG X - DIRECTION :

Mu = 2.111 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.111X10^7/(1000X207^2) = 0.493 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X207/100 = 2.69 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.051 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.051X10^7/(1000X219^2) = 0.219 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X219/100 = 1.31 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-2.586 - 0.196 =	-2.782	T-M
My1 =	My - abs(Mxy) =	-1.632 - 0.196 =	-1.828	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-2.586 - 0.024 =	-2.61	T-M
My2 =	My-abs(Mxy^2/Mx) =	-1.632 - 0.015 =	-1.617	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -2.782 T-M
Myd = -1.828 T-M

ALONG X - DIRECTION :

Mu = 2.782 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.782X10^7/(1000X207^2) = 0.649 N/mm^2 < 6.9 N/mm^2
Pt = 0.0017 Reqd. Ast = 0.0017X1000X207/100 = 3.52 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.828 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.828X10^7/(1000X219^2) = 0.381 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Outotec / L&T Revisic	0	1	2	3
Date	22-05-2020	29-06-2020	16.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.157 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 124 , L/C No. - 100)*
Vu = 3.25 T

Provided Pt = 0.0027
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0027X3.25X10^4X207/2.782/10^7)X1000X207/10^4
= 19.824 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4
= 20.544 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X207/10^4
= 35.514 T
HENCE, Vc = 19.824 T
Φt*Vc = 0.75X19.824 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 14.868 T **> 3.25T , Hence O.K.**

ALONG Y - DIRECTION :

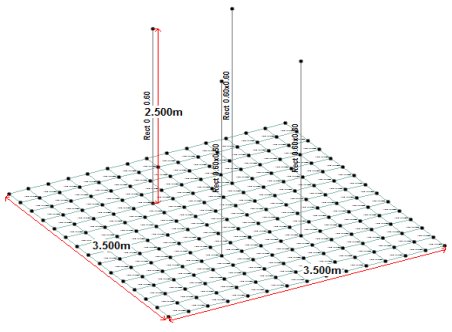
SQy = -0.069 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 129 , L/C No. - 100)*
Vu = 1.511 T

Pt provided = 0.0026
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X1.511X10^4X219/1.828/10^7)X1000X219/10^4
= 20.905 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X219/10^4
= 37.573 T
HENCE, Vc = 20.905 T
Φt*Vc = 0.75X20.905 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.679 T **> 1.511T , Hence O.K.**

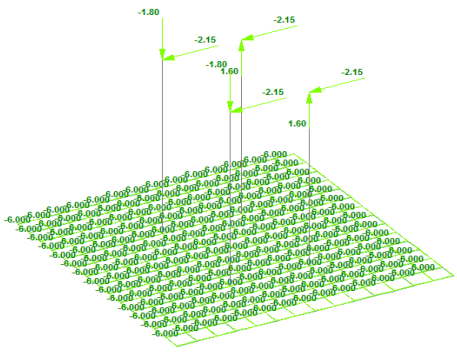
DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F9

Size of Column = 600 X 600 mm^2
Reqired area of Reinforcement = 3600 mm^2
Reinforcement Provided = 8 no. 25 dia bars
Area of Reinforcement Provided = 3927 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 2 T/m^2
Soil Weight over the foundation = 2.2 x 1.8 T/m2 = 3.96 T/m^2
SOIL+SURCHARGE LOAD = 5.96 T/m^2

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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revisio	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 STAAD Input & Output File of Foundation Mkd. F2

```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 22, 2020                   *
*           Time=    10:22:46                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDN_F2.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 -4.5; 3 0 2.5 0; 4 0 2.5 -4.5; 5 -1 0 0; 7 1 0 0; 9 -0.75 0 0
     9. 10 -0.5 0 0; 11 -0.25 0 0; 12 0.25 0 0; 13 0.5 0 0; 14 0.75 0 0; 15 0 0 -0.25
    10. 16 -1 0 -0.25; 17 1 0 -0.25; 18 -0.75 0 -0.25; 19 -0.5 0 -0.25
    11. 20 -0.25 0 -0.25; 21 0.25 0 -0.25; 22 0.5 0 -0.25; 23 0.75 0 -0.25
    12. 24 0 0 -0.5; 25 -1 0 -0.5; 26 1 0 -0.5; 27 -0.75 0 -0.5; 28 -0.5 0 -0.5
    13. 29 -0.25 0 -0.5; 30 0.25 0 -0.5; 31 0.5 0 -0.5; 32 0.75 0 -0.5; 33 0 0 -0.75
    14. 34 -1 0 -0.75; 35 1 0 -0.75; 36 -0.75 0 -0.75; 37 -0.5 0 -0.75
    15. 38 -0.25 0 -0.75; 39 0.25 0 -0.75; 40 0.5 0 -0.75; 41 0.75 0 -0.75; 42 0 0 -1.
    16. 43 -1 0 -1; 44 1 0 -1; 45 -0.75 0 -1; 46 -0.5 0 -1; 47 -0.25 0 -1.
    17. 48 0.25 0 -1; 49 0.5 0 -1; 50 0.75 0 -1; 51 0 0 -1.25; 52 -1 0 -1.25
    18. 53 1 0 -1.25; 54 -0.75 0 -1.25; 55 -0.5 0 -1.25; 56 -0.25 0 -1.25
    19. 57 0.25 0 -1.25; 58 0.5 0 -1.25; 59 0.75 0 -1.25; 60 0 0 -1.5; 61 -1 0 -1.5
    20. 62 1 0 -1.5; 63 -0.75 0 -1.5; 64 -0.5 0 -1.5; 65 -0.25 0 -1.5; 66 0.25 0 -1.5
    21. 67 0.5 0 -1.5; 68 0.75 0 -1.5; 69 0 0 -1.75; 70 -1 0 -1.75; 71 1 0 -1.75
    22. 72 -0.75 0 -1.75; 73 -0.5 0 -1.75; 74 -0.25 0 -1.75; 75 0.25 0 -1.75
    23. 76 0.5 0 -1.75; 77 0.75 0 -1.75; 78 0 0 -2; 79 -1 0 -2; 80 1 0 -2
    24. 81 -0.75 0 -2; 82 -0.5 0 -2; 83 -0.25 0 -2; 84 0.25 0 -2; 85 0.5 0 -2
    25. 86 0.75 0 -2; 87 0 0 -2.25; 88 -1 0 -2.25; 89 1 0 -2.25; 90 -0.75 0 -2.25
    26. 91 -0.5 0 -2.25; 92 -0.25 0 -2.25; 93 0.25 0 -2.25; 94 0.5 0 -2.25
    27. 95 0.75 0 -2.25; 96 0 0 -2.5; 97 -1 0 -2.5; 98 1 0 -2.5; 99 -0.75 0 -2.5
    28. 100 -0.5 0 -2.5; 101 -0.25 0 -2.5; 102 0.25 0 -2.5; 103 0.5 0 -2.5
    29. 104 0.75 0 -2.5; 105 0 0 -2.75; 106 -1 0 -2.75; 107 1 0 -2.75
    30. 108 -0.75 0 -2.75; 109 -0.5 0 -2.75; 110 -0.25 0 -2.75; 111 0.25 0 -2.75
    31. 112 0.5 0 -2.75; 113 0.75 0 -2.75; 114 0 0 -3; 115 -1 0 -3; 116 1 0 -3
    32. 117 -0.75 0 -3; 118 -0.5 0 -3; 119 -0.25 0 -3; 120 0.25 0 -3; 121 0.5 0 -3
    33. 122 0.75 0 -3; 123 0 0 -3.25; 124 -1 0 -3.25; 125 1 0 -3.25; 126 -0.75 0 -3.25
    34. 127 -0.5 0 -3.25; 128 -0.25 0 -3.25; 129 0.25 0 -3.25; 130 0.5 0 -3.25
    35. 131 0.75 0 -3.25; 132 0 0 -3.5; 133 -1 0 -3.5; 134 1 0 -3.5; 135 -0.75 0 -3.5
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    38. 144 -0.75 0 -3.75; 145 -0.5 0 -3.75; 146 -0.25 0 -3.75; 147 0.25 0 -3.75
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39. 148 0.5 0 -3.75; 149 0.75 0 -3.75; 150 0 0 -4; 151 -1 0 -4; 152 1 0 -4
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42. 163 -0.5 0 -4.25; 164 -0.25 0 -4.25; 165 0.25 0 -4.25; 166 0.5 0 -4.25
43. 167 0.75 0 -4.25; 246 -1 0 0.25; 247 -0.75 0 0.25; 248 -0.5 0 0.25
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45. 253 0.75 0 0.25; 254 1 0 0.25; 255 -0.75 0 -4.5; 256 -1 0 -4.5
46. 257 -0.5 0 -4.5; 258 -0.25 0 -4.5; 260 0.25 0 -4.5; 261 0.5 0 -4.5
47. 262 0.75 0 -4.5; 263 1 0 -4.5; 264 -0.75 0 -4.75; 265 -1 0 -4.75
48. 266 -0.5 0 -4.75; 267 -0.25 0 -4.75; 268 0 0 -4.75; 269 0.25 0 -4.75
49. 270 0.5 0 -4.75; 271 0.75 0 -4.75; 272 1 0 -4.75; 273 -0.75 0 -5; 274 -1 0 -5
50. 275 -0.5 0 -5; 276 -0.25 0 -5; 277 0 0 -5; 278 0.25 0 -5; 279 0.5 0 -5
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54. 295 1.25 0 -1; 296 1.25 0 -1.25; 297 1.25 0 -1.5; 298 1.25 0 -1.75
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56. 303 1.25 0 -3; 304 1.25 0 -3.25; 305 1.25 0 -3.5; 306 1.25 0 -3.75
57. 307 1.25 0 -4; 308 1.25 0 -4.25; 309 1.25 0 0.25; 310 1.25 0 -4.5
58. 311 1.25 0 -4.75; 312 1.25 0 -5; 313 1.25 0 0.5; 314 -1.25 0 0
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60. 319 -1.25 0 -1.25; 320 -1.25 0 -1.5; 321 -1.25 0 -1.75; 322 -1.25 0 -2
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62. 327 -1.25 0 -3.25; 328 -1.25 0 -3.5; 329 -1.25 0 -3.75; 330 -1.25 0 -4
63. 331 -1.25 0 -4.25; 332 -1.25 0 0.25; 333 -1.25 0 -4.5; 334 -1.25 0 -4.75
64. 335 -1.25 0 -5; 336 -1.25 0 0.5
65. MEMBER INCIDENCES
66. 1 1 3; 318 2 4
67. ELEMENT INCIDENCES SHELL
68. 17 5 9 18 16; 18 9 10 19 18; 19 10 11 20 19; 20 11 1 15 20; 21 1 12 21 15
69. 22 12 13 22 21; 23 13 14 23 22; 24 14 7 17 23; 28 16 18 27 25; 29 18 19 28 27
70. 30 19 20 29 28; 31 20 15 24 29; 32 15 21 30 24; 33 21 22 31 30; 34 22 23 32 31
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89. 143 111 112 121 120; 144 112 113 122 121; 145 113 107 116 122
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92. 155 121 122 131 130; 156 122 116 125 131; 160 124 126 135 133
93. 161 126 127 136 135; 162 127 128 137 136; 163 128 123 132 137
94. 164 123 129 138 132; 165 129 130 139 138; 166 130 131 140 139

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 95. 167 131 125 134 140; 171 133 135 144 142; 172 135 136 145 144
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 97. 176 138 139 148 147; 177 139 140 149 148; 178 140 134 143 149
 98. 182 142 144 153 151; 183 144 145 154 153; 184 145 146 155 154
 99. 185 146 141 150 155; 186 141 147 156 150; 187 147 148 157 156
100. 188 148 149 158 157; 189 149 143 152 158; 193 151 153 162 160
101. 194 153 154 163 162; 195 154 155 164 163; 196 155 150 159 164
102. 197 150 156 165 159; 198 156 157 166 165; 199 157 158 167 166
103. 200 158 152 161 167; 294 246 247 9 5; 295 247 248 10 9; 296 248 249 11 10
104. 297 249 250 1 11; 298 250 251 12 1; 299 251 252 13 12; 300 252 253 14 13
105. 301 253 254 7 14; 302 160 162 255 256; 303 162 163 257 255
106. 304 163 164 258 257; 305 164 159 2 258; 306 159 165 260 2; 307 165 166 261 260
107. 308 166 167 262 261; 309 167 161 263 262; 310 256 255 264 265
108. 311 255 257 266 264; 312 257 258 267 266; 313 258 2 268 267; 314 2 260 269 268
109. 315 260 261 270 269; 316 261 262 271 270; 317 262 263 272 271
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112. 325 270 271 280 279; 326 271 272 281 280; 327 282 283 247 246
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114. 331 286 287 251 250; 332 287 288 252 251; 333 288 289 253 252
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117. 342 71 298 299 80; 343 80 299 300 89; 344 89 300 301 98; 345 98 301 302 107
118. 346 107 302 303 116; 347 116 303 304 125; 348 125 304 305 134
119. 349 134 305 306 143; 350 143 306 307 152; 351 152 307 308 161
120. 352 254 309 291 7; 353 161 308 310 263; 354 263 310 311 272
121. 355 272 311 312 281; 356 290 313 309 254; 357 314 5 16 315; 358 315 16 25 316
122. 359 316 25 34 317; 360 317 34 43 318; 361 318 43 52 319; 362 319 52 61 320
123. 363 320 61 70 321; 364 321 70 79 322; 365 322 79 88 323; 366 323 88 97 324
124. 367 324 97 106 325; 368 325 106 115 326; 369 326 115 124 327
125. 370 327 124 133 328; 371 328 133 142 329; 372 329 142 151 330
126. 373 330 151 160 331; 374 332 246 5 314; 375 331 160 256 333
127. 376 333 256 265 334; 377 334 265 274 335; 378 336 282 246 332
128. START GROUP DEFINITION
129. ELEMENT
130. _BM 17 TO 19 22 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 -
131. 94 TO 101 105 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 -
132. 171 TO 178 182 TO 189 193 TO 200 294 TO 296 299 TO 304 307 TO 312 -
133. 315 TO 317 319 TO 378
134. _SF 17 18 23 24 28 29 34 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 -
135. 95 TO 101 105 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 -
136. 171 TO 178 182 TO 189 193 194 199 200 294 295 300 TO 303 308 TO 311 316 317 -
137. 319 320 325 TO 328 333 TO 378
138. END GROUP DEFINITION
139. ELEMENT PROPERTY
140. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
141. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
142. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 THICKNESS 0.35
143. DEFINE MATERIAL START
144. ISOTROPIC CONCRETE
145. E 2.21467E+006
146. POISSON 0.17
147. DENSITY 2.40262
148. ALPHA 1E-005
149. DAMP 0.05
150. TYPE CONCRETE
```

```
151. STRENGTH FCU 2812.28
152. END DEFINE MATERIAL
153. MEMBER PROPERTY AMERICAN
154. 1 318 PRIS YD 0.6 ZD 0.6
155. CONSTANTS
156. MATERIAL CONCRETE ALL
157. SUPPORTS
158. 1 2 5 7 9 TO 167 246 TO 258 260 TO 336 ELASTIC MAT DIRECT Y SUBGRADE 4500 PRINT
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JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
5	0.062500
7	0.062500
9	0.062500
10	0.062500
11	0.062500
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272	0.062500
273	0.031250
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306	0.031250
307	0.031250
308	0.031250
309	0.031250
310	0.031250
311	0.031250
312	0.015625
313	0.015625
314	0.031250
315	0.031250
316	0.031250
317	0.031250
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326	0.031250
327	0.031250
328	0.031250
329	0.031250
330	0.031250
331	0.031250
332	0.031250
333	0.031250
334	0.031250
335	0.015625
336	0.015625
TOTAL	13.750000

```
159. LOAD 1 DL+LL
160. *SOIL+SURCHARGE=1.65X1.8+2=5T/SQ.M)
161. SELFWEIGHT Y -1
162. ELEMENT LOAD
163. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
164. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
165. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5
166. JOINT LOAD
167. 4 FX 0.15 FY -14.12 FZ -0.02
168. 3 FX 0.35 FY -14.63 FZ 0.08
169. LOAD 2 DL+0.75*LL+0.75*WLX
170. SELFWEIGHT Y -1
171. ELEMENT LOAD
172. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
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173. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
174. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5
175. JOINT LOAD
176. 4 FX 2.89 FY -17.48 FZ -0.05
177. 3 FX 3.32 FY -18.52 FZ 0.05
178. LOAD 3 DL+0.75*LL+0.75*WLZ
179. SELFWEIGHT Y -1
180. ELEMENT LOAD
181. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
182. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
183. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5
184. JOINT LOAD
185. 4 FX 0.03 FY -3.86 FZ 2.65
186. 3 FX 0.46 FY -20.71 FZ 2.75
187. LOAD 4 0.6*DL+WLX
188. *SOIL+SURCHARGE=1.65X1.8+0.6*2=4.2T/SQ.M)
189. SELFWEIGHT Y -1
190. ELEMENT LOAD
191. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
192. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
193. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -4.2
194. JOINT LOAD
195. 4 FX 3.74 FY -10.15 FZ -0.04
196. 3 FX 4.16 FY -11.21 FZ 0
197. LOAD 5 0.6*DL+WLZ
198. SELFWEIGHT Y -1
199. ELEMENT LOAD
200. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
201. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
202. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -4.2
203. JOINT LOAD
204. 4 FX 0.07 FY 8.01 FZ 3.56
205. 3 FX 0.34 FY -14.14 FZ 3.6
206. ***********************************************
207. LOAD 100 1.4*DL
208. *SOIL+SURCHARGE=1.65X1.8+1.4*2=5.8T/SQ.M)
209. SELFWEIGHT Y -1
210. ELEMENT LOAD
211. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
212. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
213. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5.8
214. JOINT LOAD
215. 4 FX 0.01 FY -5.28 FZ 0.02
216. 3 FX 0.45 FY -6.06 FZ 0.1
217. LOAD 101 1.4*DL + 1.7*LL
218. SELFWEIGHT Y -1
219. ELEMENT LOAD
220. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
221. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
222. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5.8
223. JOINT LOAD
224. 4 FX 0.22 FY -22.92 FZ -0.04
225. 3 FX 0.5 FY -23.58 FZ 0.11
226. LOAD 102 1.2*DL + 1.6*WLX + LL
227. *SOIL+SURCHARGE=1.65X1.8+1.2*2=5.4T/SQ.M)
228. SELFWEIGHT Y -1
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229. ELEMENT LOAD
230. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
231. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
232. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5.4
233. JOINT LOAD
234. 4 FX 6.04 FY -27.61 FZ 0.09
235. 3 FX 0.03 FY -29.28
236. LOAD 103 1.2*DL + 1.6*WLZ + LL
237. SELFWEIGHT Y -1
238. ELEMENT LOAD
239. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
240. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
241. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -5.4
242. JOINT LOAD
243. 4 FX -0.07 FY 1.45 FZ 5.66
244. 3 FX 0.65 FY -33.97 FZ 5.79
245. LOAD 104 0.9*DL + 1.6*WLX
246. *SOIL+SURCHARGE=1.65X1.8+0.6*2=4.8T/SQ.M)
247. SELFWEIGHT Y -1
248. ELEMENT LOAD
249. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
250. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
251. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -4.8
252. JOINT LOAD
253. 4 FX 5.98 FY -16.03 FZ -0.06
254. 3 FX 6.64 FY -17.67 FZ 0
255. LOAD 105 0.9*DL + 1.6*WLZ
256. SELFWEIGHT Y -1
257. ELEMENT LOAD
258. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
259. 106 TO 112 116 TO 123 127 TO 134 138 TO 145 149 TO 156 160 TO 167 171 TO 178 -
260. 182 TO 189 193 TO 200 294 TO 317 319 TO 378 PR -4.8
261. JOINT LOAD
262. 4 FX -0.12 FY 13.03 FZ 5.69
263. 3 FX 0.53 FY -22.36 FZ 5.76
264. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 255 NUMBER OF MEMBERS 2
NUMBER OF PLATES 220 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 253

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 246/ 19/ 60 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 771
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 47 DOUBLE KILO-WORDS

STAAD SPACE **-- PAGE NO. 11**

REQRD/AVAIL. DISK SPACE = 13.8/ 113647.8 MB

265. START CONCRETE DESIGN

```
266. CODE ACI
267. CLS 0.0075 ALL
268. DESIGN COLUMN 1 318
```

==

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 318 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

269. END CONCRETE DESIGN
270. FINISH

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revision No.:	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 2 STAAD Input & Output File of Foundation Mkd. F4

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 22, 2020                *
*           Time=    12: 1:21                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDN_F4.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 3 0 2.5 0; 4 2 2.5 0; 5 0 0 -1; 7 0 0 1; 9 0 0 -0.75; 10 0 0 -0.5
     9. 11 0 0 -0.25; 12 0 0 0.25; 13 0 0 0.5; 14 0 0 0.75; 15 0.25 0 0; 16 0.25 0 -1.
    10. 17 0.25 0 1; 18 0.25 0 -0.75; 19 0.25 0 -0.5; 20 0.25 0 -0.25; 21 0.25 0 0.25
    11. 22 0.25 0 0.5; 23 0.25 0 0.75; 24 0.5 0 0; 25 0.5 0 -1; 26 0.5 0 1
    12. 27 0.5 0 -0.75; 28 0.5 0 -0.5; 29 0.5 0 -0.25; 30 0.5 0 0.25; 31 0.5 0 0.5
    13. 32 0.5 0 0.75; 33 0.75 0 0; 34 0.75 0 -1; 35 0.75 0 1; 36 0.75 0 -0.75
    14. 37 0.75 0 -0.5; 38 0.75 0 -0.25; 39 0.75 0 0.25; 40 0.75 0 0.5; 41 0.75 0 0.75
    15. 42 1 0 0; 43 1 0 -1; 44 1 0 1; 45 1 0 -0.75; 46 1 0 -0.5; 47 1 0 -0.25
    16. 48 1 0 0.25; 49 1 0 0.5; 50 1 0 0.75; 51 1.25 0 0; 52 1.25 0 -1; 53 1.25 0 1
    17. 54 1.25 0 -0.75; 55 1.25 0 -0.5; 56 1.25 0 -0.25; 57 1.25 0 0.25
    18. 58 1.25 0 0.5; 59 1.25 0 0.75; 60 1.5 0 0; 61 1.5 0 -1; 62 1.5 0 1
    19. 63 1.5 0 -0.75; 64 1.5 0 -0.5; 65 1.5 0 -0.25; 66 1.5 0 0.25; 67 1.5 0 0.5
    20. 68 1.5 0 0.75; 69 1.75 0 0; 70 1.75 0 -1; 71 1.75 0 1; 72 1.75 0 -0.75
    21. 73 1.75 0 -0.5; 74 1.75 0 -0.25; 75 1.75 0 0.25; 76 1.75 0 0.5; 77 1.75 0 0.75
    22. 78 2 0 0; 79 2 0 -1; 80 2 0 1; 81 2 0 -0.75; 82 2 0 -0.5; 83 2 0 -0.25
    23. 84 2 0 0.25; 85 2 0 0.5; 86 2 0 0.75; 87 2.25 0 0; 88 2.25 0 -1; 89 2.25 0 1
    24. 90 2.25 0 -0.75; 91 2.25 0 -0.5; 92 2.25 0 -0.25; 93 2.25 0 0.25
    25. 94 2.25 0 0.5; 95 2.25 0 0.75; 246 -0.25 0 -1; 247 -0.25 0 -0.75
    26. 248 -0.25 0 -0.5; 249 -0.25 0 -0.25; 250 -0.25 0 0; 251 -0.25 0 0.25
    27. 252 -0.25 0 0.5; 253 -0.25 0 0.75; 254 -0.25 0 1; 255 -0.5 0 -1.
    28. 256 -0.5 0 -0.75; 257 -0.5 0 -0.5; 258 -0.5 0 -0.25; 259 -0.5 0 0
    29. 260 -0.5 0 0.25; 261 -0.5 0 0.5; 262 -0.5 0 0.75; 263 -0.5 0 1
    30. 264 2.5 0 -0.75; 265 2.5 0 -1; 266 2.5 0 -0.5; 267 2.5 0 -0.25; 268 2.5 0 0
    31. 269 2.5 0 0.25; 270 2.5 0 0.5; 271 2.5 0 0.75; 272 2.5 0 1; 273 -0.75 0 -1.
    32. 274 -0.75 0 -0.75; 275 -0.75 0 -0.5; 276 -0.75 0 -0.25; 277 -0.75 0 0
    33. 278 -0.75 0 0.25; 279 -0.75 0 0.5; 280 -0.75 0 0.75; 281 -0.75 0 1
    34. 282 2.75 0 -0.75; 283 2.75 0 -1; 284 2.75 0 -0.5; 285 2.75 0 -0.25
    35. 286 2.75 0 0; 287 2.75 0 0.25; 288 2.75 0 0.5; 289 2.75 0 0.75; 290 2.75 0 1
    36. MEMBER INCIDENCES
    37. 1 1 3; 318 78 4
    38. ELEMENT INCIDENCES SHELL
```

```
39. 17 5 9 18 16; 18 9 10 19 18; 19 10 11 20 19; 20 11 1 15 20; 21 1 12 21 15
40. 22 12 13 22 21; 23 13 14 23 22; 24 14 7 17 23; 28 16 18 27 25; 29 18 19 28 27
41. 30 19 20 29 28; 31 20 15 24 29; 32 15 21 30 24; 33 21 22 31 30; 34 22 23 32 31
42. 35 23 17 26 32; 39 25 27 36 34; 40 27 28 37 36; 41 28 29 38 37; 42 29 24 33 38
43. 43 24 30 39 33; 44 30 31 40 39; 45 31 32 41 40; 46 32 26 35 41; 50 34 36 45 43
44. 51 36 37 46 45; 52 37 38 47 46; 53 38 33 42 47; 54 33 39 48 42; 55 39 40 49 48
45. 56 40 41 50 49; 57 41 35 44 50; 61 43 45 54 52; 62 45 46 55 54; 63 46 47 56 55
46. 64 47 42 51 56; 65 42 48 57 51; 66 48 49 58 57; 67 49 50 59 58; 68 50 44 53 59
47. 72 52 54 63 61; 73 54 55 64 63; 74 55 56 65 64; 75 56 51 60 65; 76 51 57 66 60
48. 77 57 58 67 66; 78 58 59 68 67; 79 59 53 62 68; 83 61 63 72 70; 84 63 64 73 72
49. 85 64 65 74 73; 86 65 60 69 74; 87 60 66 75 69; 88 66 67 76 75; 89 67 68 77 76
50. 90 68 62 71 77; 94 70 72 81 79; 95 72 73 82 81; 96 73 74 83 82; 97 74 69 78 83
51. 98 69 75 84 78; 99 75 76 85 84; 100 76 77 86 85; 101 77 71 80 86
52. 105 79 81 90 88; 106 81 82 91 90; 107 82 83 92 91; 108 83 78 87 92
53. 109 78 84 93 87; 110 84 85 94 93; 111 85 86 95 94; 112 86 80 89 95
54. 294 246 247 9 5; 295 247 248 10 9; 296 248 249 11 10; 297 249 250 1 11
55. 298 250 251 12 1; 299 251 252 13 12; 300 252 253 14 13; 301 253 254 7 14
56. 319 255 256 247 246; 320 256 257 248 247; 321 257 258 249 248
57. 322 258 259 250 249; 323 259 260 251 250; 324 260 261 252 251
58. 325 261 262 253 252; 326 262 263 254 253; 327 88 90 264 265; 328 90 91 266 264
59. 329 91 92 267 266; 330 92 87 268 267; 331 87 93 269 268; 332 93 94 270 269
60. 333 94 95 271 270; 334 95 89 272 271; 335 273 274 256 255; 336 274 275 257 256
61. 337 275 276 258 257; 338 276 277 259 258; 339 277 278 260 259
62. 340 278 279 261 260; 341 279 280 262 261; 342 280 281 263 262
63. 343 265 264 282 283; 344 264 266 284 282; 345 266 267 285 284
64. 346 267 268 286 285; 347 268 269 287 286; 348 269 270 288 287
65. 349 270 271 289 288; 350 271 272 290 289
66. START GROUP DEFINITION
67. ELEMENT
68. _BM 17 TO 19 22 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 -
69. 94 TO 96 99 TO 101 105 TO 107 110 TO 112 294 TO 296 299 TO 301 319 TO 350
70. _SF 17 18 23 24 28 29 34 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 84 89 90 -
71. 94 95 100 101 105 106 111 112 294 295 300 301 319 320 325 TO 328 333 TO 350
72. END GROUP DEFINITION
73. ELEMENT PROPERTY
74. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
75. 106 TO 112 294 TO 301 319 TO 350 THICKNESS 0.3
76. DEFINE MATERIAL START
77. ISOTROPIC CONCRETE
78. E 2.21467E+006
79. POISSON 0.17
80. DENSITY 2.40262
81. ALPHA 1E-005
82. DAMP 0.05
83. TYPE CONCRETE
84. STRENGTH FCU 2812.28
85. END DEFINE MATERIAL
86. MEMBER PROPERTY AMERICAN
87. 1 318 PRIS YD 0.6 ZD 0.6
88. CONSTANTS
89. MATERIAL CONCRETE ALL
90. SUPPORTS
91. 1 5 7 9 TO 95 246 TO 290 ELASTIC MAT DIRECT Y SUBGRADE 4500 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
5	0.031250
7	0.031250
9	0.062500
10	0.062500
11	0.062500
12	0.062500
13	0.062500
14	0.062500
15	0.062500
16	0.031250
17	0.031250
18	0.062500
19	0.062500
20	0.062500
21	0.062500
22	0.062500
23	0.062500
24	0.062500
25	0.031250
26	0.031250
27	0.062500
28	0.062500
29	0.062500
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.031250
35	0.031250
36	0.062500
37	0.062500
38	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.031250
44	0.031250
45	0.062500
46	0.062500
47	0.062500
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.031250
53	0.031250
54	0.062500
55	0.062500
56	0.062500
57	0.062500

58	0.062500
59	0.062500
60	0.062500
61	0.031250
62	0.031250
63	0.062500
64	0.062500
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.031250
71	0.031250
72	0.062500
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.031250
80	0.031250
81	0.062500
82	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.031250
89	0.031250
90	0.062500
91	0.062500
92	0.062500
93	0.062500
94	0.062500
95	0.062500
246	0.031250
247	0.062500
248	0.062500
249	0.062500
250	0.062500
251	0.062500
252	0.062500
253	0.062500
254	0.031250
255	0.031250
256	0.062500
257	0.062500
258	0.062500
259	0.062500
260	0.062500
261	0.062500
262	0.062500
263	0.031250

264	0.062500
265	0.031250
266	0.062500
267	0.062500
268	0.062500
269	0.062500
270	0.062500
271	0.062500
272	0.031250
273	0.015625
274	0.031250
275	0.031250
276	0.031250
277	0.031250
278	0.031250
279	0.031250
280	0.031250
281	0.015625
282	0.031250
283	0.015625
284	0.031250
285	0.031250
286	0.031250
287	0.031250
288	0.031250
289	0.031250
290	0.015625
TOTAL	7.000000

```
 92. LOAD 1 DL+LL
 93. *SOIL+SURCHARGE=1.7X1.8+2=5T/SQ.M)
 94. SELFWEIGHT Y -1
 95. ELEMENT LOAD
 96. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
 97. 106 TO 112 294 TO 301 319 TO 350 PR -5
 98. JOINT LOAD
 99. 3 FX 0.01 FY -4.96 FZ 0
100. 4 FX 0.11 FY -1.66 FZ 0
101. LOAD 2 DL+0.75*LL+0.75*WLX
102. SELFWEIGHT Y -1
103. ELEMENT LOAD
104. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
105. 106 TO 112 294 TO 301 319 TO 350 PR -5
106. JOINT LOAD
107. 3 FX 0 FY 2.38 FZ 0
108. 4 FX 2.24 FY -9 FZ 0
109. LOAD 3 DL+0.75*LL+0.75*WLZ
110. SELFWEIGHT Y -1
111. ELEMENT LOAD
112. 18 TO 23 29 TO 34 40 TO 45 51 TO 56 62 TO 67 73 TO 78 84 TO 89 95 TO 100 106 -
113. 107 TO 111 295 TO 300 320 TO 325 328 TO 333 336 TO 341 344 TO 349 PR -5
114. JOINT LOAD
115. 3 FX 0.01 FY -5.23 FZ 0.1
116. 4 FX 0.13 FY -1.39 FZ 0.12
117. LOAD 4 0.6*DL+WLX
118. *SOIL+SURCHARGE=1.7X1.8+0.6*2=4.3T/SQ.M
119. SELFWEIGHT Y -1
```

```
120. ELEMENT LOAD
121. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
122. 106 TO 112 294 TO 301 319 TO 350 PR -4.3
123. JOINT LOAD
124. 3 FX 0 FY 6.8 FZ 0
125. 4 FX 2.9 FY -10.78 FZ 0
126. LOAD 5 0.6*DL+WLZ
127. SELFWEIGHT Y -1
128. ELEMENT LOAD
129. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
130. 106 TO 112 294 TO 301 319 TO 350 PR -4.3
131. JOINT LOAD
132. 3 FX 0.01 FY -3.34 FZ 0.13
133. 4 FX 0.09 FY -0.63 FZ 0.16
134. **********************************************
135. LOAD 100 1.4*DL
136. *SOIL+SURCHARGE=1.7X1.8+1.4*2=5.86T/SQ.M
137. SELFWEIGHT Y -1
138. ELEMENT LOAD
139. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
140. 106 TO 112 294 TO 301 319 TO 350 PR -5.9
141. JOINT LOAD
142. 3 FX 0.01 FY -6.98 FZ 0
143. 4 FX 0.15 FY -2.29 FZ 0
144. LOAD 101 1.4*DL + 1.7*LL
145. SELFWEIGHT Y -1
146. ELEMENT LOAD
147. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
148. 106 TO 112 294 TO 301 319 TO 350 PR -5.9
149. JOINT LOAD
150. 3 FX 0.01 FY -6.94 FZ 0
151. 4 FX 0.16 FY -2.33 FZ 0
152. LOAD 102 1.2*DL + 1.6*WLX + LL
153. *SOIL+SURCHARGE=1.7X1.8+1.2*2=5.65T/SQ.M
154. ELEMENT LOAD
155. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
156. 106 TO 112 294 TO 301 319 TO 350 PR -5.65
157. JOINT LOAD
158. 3 FX -0.01 FY 9.71 FZ 0
159. 4 FX -4.67 FY -17.66 FZ 0
160. LOAD 103 1.2*DL + 1.6*WLZ + LL
161. ELEMENT LOAD
162. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
163. 106 TO 112 294 TO 301 319 TO 350 PR -5.65
164. JOINT LOAD
165. 3 FX 0.01 FY -6.52 FZ 0.21
166. 4 FX 0.18 FY -1.43 FZ 0.26
167. LOAD 104 0.9*DL + 1.6*WLX
168. *SOIL+SURCHARGE=1.7X1.8+0.9*2=5T/SQ.M
169. ELEMENT LOAD
170. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
171. 106 TO 112 294 TO 301 319 TO 350 PR -4.8
172. JOINT LOAD
173. 3 FX 0.01 FY 11.18 FZ 0
174. 4 FX 4.63 FY -17.15 FZ 0
175. LOAD 105 0.9*DL + 1.6*WLZ
```

```
176. ELEMENT LOAD
177. 17 TO 24 28 TO 35 39 TO 46 50 TO 57 61 TO 68 72 TO 79 83 TO 90 94 TO 101 105 -
178. 106 TO 112 294 TO 301 319 TO 350 PR -4.8
179. JOINT LOAD
180. 3 FX 0.01 FY -5.05 FZ 0.21
181. 4 FX 0.14 FY -0.91 FZ 0.26
182. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**137**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**112**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**135**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 97/ 15/ 48 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 417
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 21 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 113613.8 MB

```
183. START CONCRETE DESIGN
```

```
184. CODE ACI
185. CLS 0.075 ALL
186. DESIGN COLUMN 1 318
```

==

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 318 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

187. END CONCRETE DESIGN
188. FINISH

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	0		
Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI		

ATTACHMENT 3 STAAD Input & Output File of Foundation Mkd. F5

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 22, 2020                *
*           Time=    15:47:15                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDN_F5.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 -4.5 0 0; 3 -4.5 2 0; 4 1.25 2 0; 5 0 0 1; 9 0 0 0.75; 10 0 0 0.5
     9. 11 0 0 0.25; 12 0 0 -0.25; 13 0 0 -0.5; 14 0 0 -0.75; 15 -0.25 0 0
    10. 16 -0.25 0 1; 18 -0.25 0 0.75; 19 -0.25 0 0.5; 20 -0.25 0 0.25
    11. 21 -0.25 0 -0.25; 22 -0.25 0 -0.5; 23 -0.25 0 -0.75; 24 -0.5 0 0; 25 -0.5 0 1
    12. 27 -0.5 0 0.75; 28 -0.5 0 0.5; 29 -0.5 0 0.25; 30 -0.5 0 -0.25; 31 -0.5 0 -0.5
    13. 32 -0.5 0 -0.75; 33 -0.75 0 0; 34 -0.75 0 1; 36 -0.75 0 0.75; 37 -0.75 0 0.5
    14. 38 -0.75 0 0.25; 39 -0.75 0 -0.25; 40 -0.75 0 -0.5; 41 -0.75 0 -0.75
    15. 42 -1 0 0; 43 -1 0 1; 45 -1 0 0.75; 46 -1 0 0.5; 47 -1 0 0.25; 48 -1 0 -0.25
    16. 49 -1 0 -0.5; 50 -1 0 -0.75; 51 -1.25 0 0; 52 -1.25 0 1; 54 -1.25 0 0.75
    17. 55 -1.25 0 0.5; 56 -1.25 0 0.25; 57 -1.25 0 -0.25; 58 -1.25 0 -0.5
    18. 59 -1.25 0 -0.75; 60 -1.5 0 0; 61 -1.5 0 1; 63 -1.5 0 0.75; 64 -1.5 0 0.5
    19. 65 -1.5 0 0.25; 66 -1.5 0 -0.25; 67 -1.5 0 -0.5; 68 -1.5 0 -0.75; 69 -1.75 0 0
    20. 70 -1.75 0 1; 72 -1.75 0 0.75; 73 -1.75 0 0.5; 74 -1.75 0 0.25
    21. 75 -1.75 0 -0.25; 76 -1.75 0 -0.5; 77 -1.75 0 -0.75; 78 -2 0 0; 79 -2 0 1
    22. 81 -2 0 0.75; 82 -2 0 0.5; 83 -2 0 0.25; 84 -2 0 -0.25; 85 -2 0 -0.5
    23. 86 -2 0 -0.75; 87 -2.25 0 0; 88 -2.25 0 1; 90 -2.25 0 0.75; 91 -2.25 0 0.5
    24. 92 -2.25 0 0.25; 93 -2.25 0 -0.25; 94 -2.25 0 -0.5; 95 -2.25 0 -0.75
    25. 96 -2.5 0 0; 97 -2.5 0 1; 99 -2.5 0 0.75; 100 -2.5 0 0.5; 101 -2.5 0 0.25
    26. 102 -2.5 0 -0.25; 103 -2.5 0 -0.5; 104 -2.5 0 -0.75; 105 -2.75 0 0
    27. 106 -2.75 0 1; 108 -2.75 0 0.75; 109 -2.75 0 0.5; 110 -2.75 0 0.25
    28. 111 -2.75 0 -0.25; 112 -2.75 0 -0.5; 113 -2.75 0 -0.75; 114 -3 0 0; 115 -3 0 1
    29. 117 -3 0 0.75; 118 -3 0 0.5; 119 -3 0 0.25; 120 -3 0 -0.25; 121 -3 0 -0.5
    30. 122 -3 0 -0.75; 123 -3.25 0 0; 124 -3.25 0 1; 126 -3.25 0 0.75
    31. 127 -3.25 0 0.5; 128 -3.25 0 0.25; 129 -3.25 0 -0.25; 130 -3.25 0 -0.5
    32. 131 -3.25 0 -0.75; 132 -3.5 0 0; 133 -3.5 0 1; 135 -3.5 0 0.75; 136 -3.5 0 0.5
    33. 137 -3.5 0 0.25; 138 -3.5 0 -0.25; 139 -3.5 0 -0.5; 140 -3.5 0 -0.75
    34. 141 -3.75 0 0; 142 -3.75 0 1; 144 -3.75 0 0.75; 145 -3.75 0 0.5
    35. 146 -3.75 0 0.25; 147 -3.75 0 -0.25; 148 -3.75 0 -0.5; 149 -3.75 0 -0.75
    36. 150 -4 0 0; 151 -4 0 1; 153 -4 0 0.75; 154 -4 0 0.5; 155 -4 0 0.25
    37. 156 -4 0 -0.25; 157 -4 0 -0.5; 158 -4 0 -0.75; 159 -4.25 0 0; 160 -4.25 0 1
    38. 162 -4.25 0 0.75; 163 -4.25 0 0.5; 164 -4.25 0 0.25; 165 -4.25 0 -0.25
```

39. 166 -4.25 0 -0.5; 167 -4.25 0 -0.75; 246 0.25 0 1; 247 0.25 0 0.75
40. 248 0.25 0 0.5; 249 0.25 0 0.25; 250 0.25 0 0; 251 0.25 0 -0.25
41. 252 0.25 0 -0.5; 253 0.25 0 -0.75; 255 -4.5 0 0.75; 256 -4.5 0 1
42. 257 -4.5 0 0.5; 258 -4.5 0 0.25; 260 -4.5 0 -0.25; 261 -4.5 0 -0.5
43. 262 -4.5 0 -0.75; 264 -4.75 0 0.75; 265 -4.75 0 1; 266 -4.75 0 0.5
44. 267 -4.75 0 0.25; 268 -4.75 0 0; 269 -4.75 0 -0.25; 270 -4.75 0 -0.5
45. 271 -4.75 0 -0.75; 273 1.25 0 0; 274 0.5 0 1; 275 0.5 0 0.75; 276 0.5 0 0.5
46. 277 0.5 0 0.25; 278 0.5 0 0; 279 0.5 0 -0.25; 280 0.5 0 -0.5; 281 0.5 0 -0.75
47. 283 0.75 0 1; 284 0.75 0 0.75; 285 0.75 0 0.5; 286 0.75 0 0.25; 287 0.75 0 0
48. 288 0.75 0 -0.25; 289 0.75 0 -0.5; 290 0.75 0 -0.75; 292 1 0 1; 293 1 0 0.75
49. 294 1 0 0.5; 295 1 0 0.25; 296 1 0 0; 297 1 0 -0.25; 298 1 0 -0.5
50. 299 1 0 -0.75; 301 1.25 0 1; 302 1.25 0 0.75; 303 1.25 0 0.5; 304 1.25 0 0.25
51. 305 1.25 0 -0.25; 306 1.25 0 -0.5; 307 1.25 0 -0.75; 309 1.5 0 1
52. 310 1.5 0 0.75; 311 1.5 0 0.5; 312 1.5 0 0.25; 313 1.5 0 0; 314 1.5 0 -0.25
53. 315 1.5 0 -0.5; 316 1.5 0 -0.75; 318 1.75 0 1; 319 1.75 0 0.75; 320 1.75 0 0.5
54. 321 1.75 0 0.25; 322 1.75 0 0; 323 1.75 0 -0.25; 324 1.75 0 -0.5
55. 325 1.75 0 -0.75; 327 -5 0 0.75; 328 -5 0 1; 329 -5 0 0.5; 330 -5 0 0.25
56. 331 -5 0 0; 332 -5 0 -0.25; 333 -5 0 -0.5; 334 -5 0 -0.75; 336 0 0 1.25
57. 337 -0.25 0 1.25; 338 -0.5 0 1.25; 339 -0.75 0 1.25; 340 -1 0 1.25
58. 341 -1.25 0 1.25; 342 -1.5 0 1.25; 343 -1.75 0 1.25; 344 -2 0 1.25
59. 345 -2.25 0 1.25; 346 -2.5 0 1.25; 347 -2.75 0 1.25; 348 -3 0 1.25
60. 349 -3.25 0 1.25; 350 -3.5 0 1.25; 351 -3.75 0 1.25; 352 -4 0 1.25
61. 353 -4.25 0 1.25; 354 0.25 0 1.25; 355 -4.5 0 1.25; 356 -4.75 0 1.25
62. 357 0.5 0 1.25; 358 0.75 0 1.25; 359 1 0 1.25; 360 1.25 0 1.25; 361 1.5 0 1.25
63. 362 1.75 0 1.25; 363 -5 0 1.25; 392 -5.125 0 0.75; 393 -5.125 0 1
64. 394 -5.125 0 0.5; 395 -5.125 0 0.25; 396 -5.125 0 0; 397 -5.125 0 -0.25
65. 398 -5.125 0 -0.5; 399 -5.125 0 -0.75; 401 -5.125 0 1.25; 403 1.875 0 1
66. 404 1.875 0 0.75; 405 1.875 0 0.5; 406 1.875 0 0.25; 407 1.875 0 0
67. 408 1.875 0 -0.25; 409 1.875 0 -0.5; 410 1.875 0 -0.75; 412 1.875 0 1.25
68. 414 -4.5 0 3.5; 415 -4.5 2 3.5; 416 1.25 0 3.5; 417 1.25 2 3.5; 418 0 0 1.5
69. 419 -0.25 0 1.5; 420 -0.5 0 1.5; 421 -0.75 0 1.5; 422 -1 0 1.5
70. 423 -1.25 0 1.5; 424 -1.5 0 1.5; 425 -1.75 0 1.5; 426 -2 0 1.5
71. 427 -2.25 0 1.5; 428 -2.5 0 1.5; 429 -2.75 0 1.5; 430 -3 0 1.5
72. 431 -3.25 0 1.5; 432 -3.5 0 1.5; 433 -3.75 0 1.5; 434 -4 0 1.5
73. 435 -4.25 0 1.5; 436 0.25 0 1.5; 437 -4.5 0 1.5; 438 -4.75 0 1.5
74. 439 0.5 0 1.5; 440 0.75 0 1.5; 441 1 0 1.5; 442 1.25 0 1.5; 443 1.5 0 1.5
75. 444 1.75 0 1.5; 445 -5 0 1.5; 446 -5.125 0 1.5; 447 1.875 0 1.5; 448 0 0 1.75
76. 449 -0.25 0 1.75; 450 -0.5 0 1.75; 451 -0.75 0 1.75; 452 -1 0 1.75
77. 453 -1.25 0 1.75; 454 -1.5 0 1.75; 455 -1.75 0 1.75; 456 -2 0 1.75
78. 457 -2.25 0 1.75; 458 -2.5 0 1.75; 459 -2.75 0 1.75; 460 -3 0 1.75
79. 461 -3.25 0 1.75; 462 -3.5 0 1.75; 463 -3.75 0 1.75; 464 -4 0 1.75
80. 465 -4.25 0 1.75; 466 0.25 0 1.75; 467 -4.5 0 1.75; 468 -4.75 0 1.75
81. 469 0.5 0 1.75; 470 0.75 0 1.75; 471 1 0 1.75; 472 1.25 0 1.75; 473 1.5 0 1.75
82. 474 1.75 0 1.75; 475 -5 0 1.75; 476 -5.125 0 1.75; 477 1.875 0 1.75; 478 0 0 2
83. 479 -0.25 0 2; 480 -0.5 0 2; 481 -0.75 0 2; 482 -1 0 2; 483 -1.25 0 2
84. 484 -1.5 0 2; 485 -1.75 0 2; 486 -2 0 2; 487 -2.25 0 2; 488 -2.5 0 2
85. 489 -2.75 0 2; 490 -3 0 2; 491 -3.25 0 2; 492 -3.5 0 2; 493 -3.75 0 2
86. 494 -4 0 2; 495 -4.25 0 2; 496 0.25 0 2; 497 -4.5 0 2; 498 -4.75 0 2
87. 499 0.5 0 2; 500 0.75 0 2; 501 1 0 2; 502 1.25 0 2; 503 1.5 0 2; 504 1.75 0 2
88. 505 -5 0 2; 506 -5.125 0 2; 507 1.875 0 2; 508 0 0 2.25; 509 -0.25 0 2.25
89. 510 -0.5 0 2.25; 511 -0.75 0 2.25; 512 -1 0 2.25; 513 -1.25 0 2.25
90. 514 -1.5 0 2.25; 515 -1.75 0 2.25; 516 -2 0 2.25; 517 -2.25 0 2.25
91. 518 -2.5 0 2.25; 519 -2.75 0 2.25; 520 -3 0 2.25; 521 -3.25 0 2.25
92. 522 -3.5 0 2.25; 523 -3.75 0 2.25; 524 -4 0 2.25; 525 -4.25 0 2.25
93. 526 0.25 0 2.25; 527 -4.5 0 2.25; 528 -4.75 0 2.25; 529 0.5 0 2.25
94. 530 0.75 0 2.25; 531 1 0 2.25; 532 1.25 0 2.25; 533 1.5 0 2.25

95. 534 1.75 0 2.25; 535 -5 0 2.25; 536 -5.125 0 2.25; 537 1.875 0 2.25
96. 538 0 0 2.5; 539 -0.25 0 2.5; 540 -0.5 0 2.5; 541 -0.75 0 2.5; 542 -1 0 2.5
97. 543 -1.25 0 2.5; 544 -1.5 0 2.5; 545 -1.75 0 2.5; 546 -2 0 2.5
98. 547 -2.25 0 2.5; 548 -2.5 0 2.5; 549 -2.75 0 2.5; 550 -3 0 2.5
99. 551 -3.25 0 2.5; 552 -3.5 0 2.5; 553 -3.75 0 2.5; 554 -4 0 2.5
100. 555 -4.25 0 2.5; 556 0.25 0 2.5; 557 -4.5 0 2.5; 558 -4.75 0 2.5
101. 559 0.5 0 2.5; 560 0.75 0 2.5; 561 1 0 2.5; 562 1.25 0 2.5; 563 1.5 0 2.5
102. 564 1.75 0 2.5; 565 -5 0 2.5; 566 -5.125 0 2.5; 567 1.875 0 2.5; 568 0 0 2.75
103. 569 -0.25 0 2.75; 570 -0.5 0 2.75; 571 -0.75 0 2.75; 572 -1 0 2.75
104. 573 -1.25 0 2.75; 574 -1.5 0 2.75; 575 -1.75 0 2.75; 576 -2 0 2.75
105. 577 -2.25 0 2.75; 578 -2.5 0 2.75; 579 -2.75 0 2.75; 580 -3 0 2.75
106. 581 -3.25 0 2.75; 582 -3.5 0 2.75; 583 -3.75 0 2.75; 584 -4 0 2.75
107. 585 -4.25 0 2.75; 586 0.25 0 2.75; 587 -4.5 0 2.75; 588 -4.75 0 2.75
108. 589 0.5 0 2.75; 590 0.75 0 2.75; 591 1 0 2.75; 592 1.25 0 2.75; 593 1.5 0 2.75
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428. 964 944 943 865 889; 965 945 944 889 913
429. START GROUP DEFINITION
430. ELEMENT

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431. _BM 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 -
432. 105 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 -
433. 171 TO 177 182 TO 188 193 TO 199 294 TO 300 302 TO 304 307 308 310 TO 312 -
434. 315 316 319 TO 325 327 TO 333 335 TO 341 343 TO 345 348 349 351 TO 353 356 -
435. 357 359 TO 365 367 TO 373 375 TO 401 433 TO 440 443 TO 450 453 TO 702 705 -
436. 706 TO 707 710 TO 731 734 TO 736 739 TO 933 935 TO 965
437. _SF 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 -
438. 105 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 -
439. 294 TO 300 375 TO 401 440 450 453 TO 611 615 627 TO 640 644 656 TO 669 673 -
440. 685 TO 698 702 714 TO 727 731 743 TO 756 760 772 TO 785 789 830 TO 843 847 -
441. 859 TO 872 876 895 TO 900 918 TO 923 935 TO 948 952
442. END GROUP DEFINITION
443. ELEMENT PROPERTY
444. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
445. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
446. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
447. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
448. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 THICKNESS 0.65
449. DEFINE MATERIAL START
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451. E 2.21467E+006
452. POISSON 0.17
453. DENSITY 2.40262
454. ALPHA 1E-005
455. DAMP 0.05
456. TYPE CONCRETE
457. STRENGTH FCU 2812.28
458. END DEFINE MATERIAL
459. MEMBER PROPERTY AMERICAN
460. 429 430 451 452 PRIS YD 0.75 ZD 0.75
461. CONSTANTS
462. MATERIAL CONCRETE ALL
463. SUPPORTS
464. 1 2 5 9 TO 16 18 TO 25 27 TO 34 36 TO 43 45 TO 52 54 TO 61 63 TO 70 -
465. 72 TO 79 81 TO 88 90 TO 97 99 TO 106 108 TO 115 117 TO 124 126 TO 133 135 -
466. 136 TO 142 144 TO 151 153 TO 160 162 TO 167 246 TO 253 255 TO 258 260 TO 262 -
467. 264 TO 271 273 TO 281 283 TO 290 292 TO 299 301 TO 307 309 TO 316 -
468. 318 TO 325 327 TO 334 336 TO 363 392 TO 399 401 403 TO 410 412 414 416 418 -
469. 419 TO 945 ELASTIC MAT DIRECT Y SUBGRADE 4500 PRINT
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775	0.031250
776	0.015625
777	0.015625
778	0.015625
779	0.015625
780	0.015625
781	0.015625
782	0.015625
783	0.015625
784	0.015625
785	0.015625
786	0.015625
787	0.015625
788	0.015625
789	0.015625
790	0.015625
791	0.015625
792	0.015625
793	0.015625
794	0.015625
795	0.015625
796	0.015625
797	0.015625
798	0.015625
799	0.015625
800	0.015625
801	0.015625
802	0.011719
803	0.011719
804	0.003906
805	0.007813
806	0.062500
807	0.062500
808	0.062500
809	0.062500
810	0.062500
811	0.062500
812	0.062500
813	0.062500
814	0.062500
815	0.062500
816	0.062500
817	0.062500
818	0.062500
819	0.062500
820	0.062500
821	0.062500
822	0.062500
823	0.062500
824	0.062500
825	0.062500
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827	0.062500
828	0.062500
829	0.062500
830	0.062500

831	0.062500
832	0.046875
833	0.046875
834	0.015625
835	0.031250
836	0.062500
837	0.062500
838	0.062500
839	0.062500
840	0.062500
841	0.062500
842	0.062500
843	0.062500
844	0.062500
845	0.062500
846	0.062500
847	0.062500
848	0.062500
849	0.062500
850	0.062500
851	0.062500
852	0.062500
853	0.062500
854	0.062500
855	0.062500
856	0.062500
857	0.062500
858	0.062500
859	0.062500
860	0.062500
861	0.062500
862	0.046875
863	0.046875
864	0.015625
865	0.031250
866	0.031250
867	0.031250
868	0.031250
869	0.031250
870	0.031250
871	0.031250
872	0.031250
873	0.031250
874	0.031250
875	0.031250
876	0.031250
877	0.031250
878	0.031250
879	0.031250
880	0.031250
881	0.031250
882	0.031250
883	0.031250
884	0.031250
885	0.023438
886	0.031250

887	0.007813
888	0.031250
889	0.031250
890	0.015625
891	0.015625
892	0.015625
893	0.015625
894	0.015625
895	0.015625
896	0.015625
897	0.015625
898	0.015625
899	0.015625
900	0.015625
901	0.015625
902	0.015625
903	0.015625
904	0.015625
905	0.015625
906	0.015625
907	0.015625
908	0.015625
909	0.011719
910	0.015625
911	0.003906
912	0.015625
913	0.015625
914	0.031250
915	0.031250
916	0.031250
917	0.031250
918	0.031250
919	0.031250
920	0.031250
921	0.031250
922	0.031250
923	0.031250
924	0.031250
925	0.031250
926	0.031250
927	0.031250
928	0.031250
929	0.031250
930	0.031250
931	0.031250
932	0.031250
933	0.031250
934	0.031250
935	0.031250
936	0.031250
937	0.031250
938	0.031250
939	0.031250
940	0.023437
941	0.023437
942	0.007812

```
       943          0.015625
       944          0.015625
       945          0.007813
    TOTAL          42.593750
470. LOAD 1 DL+LL
471. SELFWEIGHT Y -1
472. ELEMENT LOAD
473. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
474. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
475. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
476. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
477. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
478. JOINT LOAD
479. 3 FX 0.66 FY -32.4 FZ -0.13
480. 415 FX 0.14 FY -51.24 FZ 0.09
481. 4 FX 0.9 FY -29.18 FZ 0.36
482. 417 FX 0.4 FY -38.56 FZ 0.55
483. LOAD 2 DL+0.75*LL+0.75*WLX
484. SELFWEIGHT Y -1
485. ELEMENT LOAD
486. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
487. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
488. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
489. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
490. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
491. JOINT LOAD
492. 3 FX 7.04 FY -21.81 FZ -0.45
493. 415 FX 2.54 FY -50.48 FZ -0.23
494. 4 FX 7.3 FY -57.98 FZ 1.25
495. 417 FX 2.81 FY -49.59 FZ 1.5
496. LOAD 3 DL+0.75*LL+0.75*WLZ
497. SELFWEIGHT Y -1
498. ELEMENT LOAD
499. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
500. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
501. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
502. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
503. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
504. JOINT LOAD
505. 3 FX 1.46 FY -19.58 FZ 3.71
506. 415 FX 0.65 FY -77.96 FZ 4.09
507. 4 FX 1.7 FY -24.72 FZ 3.43
508. 417 FX 0.93 FY -57.97 FZ 3.78
509. LOAD 4 0.6*DL+WLX
510. SELFWEIGHT Y -1
511. ELEMENT LOAD
512. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
513. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
514. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
515. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
516. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3
517. JOINT LOAD
518. 3 FX 7.6 FY 20.12 FZ -0.45
519. 415 FX 2.95 FY -2.05 FZ -0.42
520. 4 FX 7.74 FY -32.16 FZ 0.8
521. 417 FX 3.09 FY -18.91 FZ 0.92
```

```
LOAD 5 0.6*DL+WLZ
SELFWEIGHT Y -1
ELEMENT LOAD
17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3
JOINT LOAD
3 FX 0.16 FY 23.09 FZ 5.1
415 FX 0.42 FY -38.7 FZ 5.34
4 FX 0.27 FY 12.19 FZ 3.7
417 FX 0.58 FY -30.08 FZ 3.95
*****************************************
LOAD 100 1.4*DL
SELFWEIGHT Y -1
ELEMENT LOAD
17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4.3
JOINT LOAD
3 FX -0.19 FY -17.95 FZ -0.09
415 FX 0.11 FY -24.46 FZ -0.09
4 FX 0.11 FY -15.55 FZ -0.03
417 FX 0.2 FY -19.05 FZ 0.14
LOAD 101 1.4*DL + 1.7*LL
SELFWEIGHT Y -1
ELEMENT LOAD
17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4.3
JOINT LOAD
3 FX 1.16 FY -51.23 FZ -0.2
415 FX 0.27 FY -81.87 FZ 0.14
4 FX 1.5 FY -46.27 FZ 0.62
417 FX 0.64 FY -61.47 FZ 0.91
LOAD 102 1.2*DL + 1.6*WLX + LL
SELFWEIGHT Y -1
ELEMENT LOAD
17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4
JOINT LOAD
3 FX 12.91 FY 9.54 FZ -0.8
415 FX 4.93 FY -41.24 FZ -0.62
4 FX 13.22 FY -72.19 FZ 1.65
417 FX 5.24 FY -58.48 FZ 1.95
LOAD 103 1.2*DL + 1.6*WLZ + LL
SELFWEIGHT Y -1
ELEMENT LOAD
```

```
578. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
579. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
580. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
581. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
582. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4
583. JOINT LOAD
584. 3 FX 1.01 FY 14.3 FZ 8.08
585. 415 FX 0.88 FY -99.88 FZ 8.6
586. 4 FX 1.26 FY -1.23 FZ 6.3
587. 417 FX 1.22 FY -76.36 FZ 6.81
588. LOAD 104 0.9*DL + 1.6*WLX
589. SELFWEIGHT Y -1
590. ELEMENT LOAD
591. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
592. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
593. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
594. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
595. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.5
596. JOINT LOAD
597. 3 FX 12.16 FY 32.96 FZ -0.72
598. 415 FX 4.73 FY -2.23 FZ -0.67
599. 4 FX 12.37 FY -50.79 FZ 1.27
600. 417 FX 4.94 FY -29.44 FZ 1.46
601. LOAD 105 0.9*DL + 1.6*WLZ
602. SELFWEIGHT Y -1
603. ELEMENT LOAD
604. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
605. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
606. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
607. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
608. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.5
609. JOINT LOAD
610. 3 FX 0.26 FY 37.72 FZ 8.16
611. 415 FX 0.68 FY -60.87 FZ 8.55
612. 4 FX 0.42 FY 20.17 FZ 5.92
613. 417 FX 0.92 FY -47.32 FZ -6.32
614. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**804**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**744**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**800**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 612/ 41/ 126 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 2424
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 306 DOUBLE KILO-WORDS

STAAD SPACE **-- PAGE NO. 27**

REQRD/AVAIL. DISK SPACE = 19.5/ 113603.3 MB

615. START CONCRETE DESIGN

```
616. CODE ACI
617. CLS 0.0075 ALL
618. TRACK 0 ALL
619. DESIGN COLUMN 429 430 451 452
```

==

COLUMN NO. 429 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 430 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 451 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 16 MM 1.001 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==========

COLUMN NO. 452 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

28 - 16 MM 1.001 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

620. END CONCRETE DESIGN
621. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 15:47:58 ****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revision No.:	0	1	2	3		
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 STAAD Input & Output File of Foundation Mkd. F6

```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 22, 2020                   *
*           Time=    11:25:17                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDN_F6.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 -4.5 0 0; 3 -4.5 2 0; 4 1.25 2 0; 5 0 0 1; 9 0 0 0.75; 10 0 0 0.5
     9. 11 0 0 0.25; 12 0 0 -0.25; 13 0 0 -0.5; 14 0 0 -0.75; 15 -0.25 0 0
    10. 16 -0.25 0 1; 18 -0.25 0 0.75; 19 -0.25 0 0.5; 20 -0.25 0 0.25
    11. 21 -0.25 0 -0.25; 22 -0.25 0 -0.5; 23 -0.25 0 -0.75; 24 -0.5 0 0; 25 -0.5 0 1
    12. 27 -0.5 0 0.75; 28 -0.5 0 0.5; 29 -0.5 0 0.25; 30 -0.5 0 -0.25; 31 -0.5 0 -0.5
    13. 32 -0.5 0 -0.75; 33 -0.75 0 0; 34 -0.75 0 1; 36 -0.75 0 0.75; 37 -0.75 0 0.5
    14. 38 -0.75 0 0.25; 39 -0.75 0 -0.25; 40 -0.75 0 -0.5; 41 -0.75 0 -0.75
    15. 42 -1 0 0; 43 -1 0 1; 45 -1 0 0.75; 46 -1 0 0.5; 47 -1 0 0.25; 48 -1 0 -0.25
    16. 49 -1 0 -0.5; 50 -1 0 -0.75; 51 -1.25 0 0; 52 -1.25 0 1; 54 -1.25 0 0.75
    17. 55 -1.25 0 0.5; 56 -1.25 0 0.25; 57 -1.25 0 -0.25; 58 -1.25 0 -0.5
    18. 59 -1.25 0 -0.75; 60 -1.5 0 0; 61 -1.5 0 1; 63 -1.5 0 0.75; 64 -1.5 0 0.5
    19. 65 -1.5 0 0.25; 66 -1.5 0 -0.25; 67 -1.5 0 -0.5; 68 -1.5 0 -0.75; 69 -1.75 0 0
    20. 70 -1.75 0 1; 72 -1.75 0 0.75; 73 -1.75 0 0.5; 74 -1.75 0 0.25
    21. 75 -1.75 0 -0.25; 76 -1.75 0 -0.5; 77 -1.75 0 -0.75; 78 -2 0 0; 79 -2 0 1
    22. 81 -2 0 0.75; 82 -2 0 0.5; 83 -2 0 0.25; 84 -2 0 -0.25; 85 -2 0 -0.5
    23. 86 -2 0 -0.75; 87 -2.25 0 0; 88 -2.25 0 1; 90 -2.25 0 0.75; 91 -2.25 0 0.5
    24. 92 -2.25 0 0.25; 93 -2.25 0 -0.25; 94 -2.25 0 -0.5; 95 -2.25 0 -0.75
    25. 96 -2.5 0 0; 97 -2.5 0 1; 99 -2.5 0 0.75; 100 -2.5 0 0.5; 101 -2.5 0 0.25
    26. 102 -2.5 0 -0.25; 103 -2.5 0 -0.5; 104 -2.5 0 -0.75; 105 -2.75 0 0
    27. 106 -2.75 0 1; 108 -2.75 0 0.75; 109 -2.75 0 0.5; 110 -2.75 0 0.25
    28. 111 -2.75 0 -0.25; 112 -2.75 0 -0.5; 113 -2.75 0 -0.75; 114 -3 0 0; 115 -3 0 1
    29. 117 -3 0 0.75; 118 -3 0 0.5; 119 -3 0 0.25; 120 -3 0 -0.25; 121 -3 0 -0.5
    30. 122 -3 0 -0.75; 123 -3.25 0 0; 124 -3.25 0 1; 126 -3.25 0 0.75
    31. 127 -3.25 0 0.5; 128 -3.25 0 0.25; 129 -3.25 0 -0.25; 130 -3.25 0 -0.5
    32. 131 -3.25 0 -0.75; 132 -3.5 0 0; 133 -3.5 0 1; 135 -3.5 0 0.75; 136 -3.5 0 0.5
    33. 137 -3.5 0 0.25; 138 -3.5 0 -0.25; 139 -3.5 0 -0.5; 140 -3.5 0 -0.75
    34. 141 -3.75 0 0; 142 -3.75 0 1; 144 -3.75 0 0.75; 145 -3.75 0 0.5
    35. 146 -3.75 0 0.25; 147 -3.75 0 -0.25; 148 -3.75 0 -0.5; 149 -3.75 0 -0.75
    36. 150 -4 0 0; 151 -4 0 1; 153 -4 0 0.75; 154 -4 0 0.5; 155 -4 0 0.25
    37. 156 -4 0 -0.25; 157 -4 0 -0.5; 158 -4 0 -0.75; 159 -4.25 0 0; 160 -4.25 0 1
    38. 162 -4.25 0 0.75; 163 -4.25 0 0.5; 164 -4.25 0 0.25; 165 -4.25 0 -0.25
```

39. 166 -4.25 0 -0.5; 167 -4.25 0 -0.75; 246 0.25 0 1; 247 0.25 0 0.75
40. 248 0.25 0 0.5; 249 0.25 0 0.25; 250 0.25 0 0; 251 0.25 0 -0.25
41. 252 0.25 0 -0.5; 253 0.25 0 -0.75; 255 -4.5 0 0.75; 256 -4.5 0 1
42. 257 -4.5 0 0.5; 258 -4.5 0 0.25; 260 -4.5 0 -0.25; 261 -4.5 0 -0.5
43. 262 -4.5 0 -0.75; 264 -4.75 0 0.75; 265 -4.75 0 1; 266 -4.75 0 0.5
44. 267 -4.75 0 0.25; 268 -4.75 0 0; 269 -4.75 0 -0.25; 270 -4.75 0 -0.5
45. 271 -4.75 0 -0.75; 273 1.25 0 0; 274 0.5 0 1; 275 0.5 0 0.75; 276 0.5 0 0.5
46. 277 0.5 0 0.25; 278 0.5 0 0; 279 0.5 0 -0.25; 280 0.5 0 -0.5; 281 0.5 0 -0.75
47. 283 0.75 0 1; 284 0.75 0 0.75; 285 0.75 0 0.5; 286 0.75 0 0.25; 287 0.75 0 0
48. 288 0.75 0 -0.25; 289 0.75 0 -0.5; 290 0.75 0 -0.75; 292 1 0 1; 293 1 0 0.75
49. 294 1 0 0.5; 295 1 0 0.25; 296 1 0 0; 297 1 0 -0.25; 298 1 0 -0.5
50. 299 1 0 -0.75; 301 1.25 0 1; 302 1.25 0 0.75; 303 1.25 0 0.5; 304 1.25 0 0.25
51. 305 1.25 0 -0.25; 306 1.25 0 -0.5; 307 1.25 0 -0.75; 309 1.5 0 1
52. 310 1.5 0 0.75; 311 1.5 0 0.5; 312 1.5 0 0.25; 313 1.5 0 0; 314 1.5 0 -0.25
53. 315 1.5 0 -0.5; 316 1.5 0 -0.75; 318 1.75 0 1; 319 1.75 0 0.75; 320 1.75 0 0.5
54. 321 1.75 0 0.25; 322 1.75 0 0; 323 1.75 0 -0.25; 324 1.75 0 -0.5
55. 325 1.75 0 -0.75; 327 -5 0 0.75; 328 -5 0 1; 329 -5 0 0.5; 330 -5 0 0.25
56. 331 -5 0 0; 332 -5 0 -0.25; 333 -5 0 -0.5; 334 -5 0 -0.75; 336 0 0 1.25
57. 337 -0.25 0 1.25; 338 -0.5 0 1.25; 339 -0.75 0 1.25; 340 -1 0 1.25
58. 341 -1.25 0 1.25; 342 -1.5 0 1.25; 343 -1.75 0 1.25; 344 -2 0 1.25
59. 345 -2.25 0 1.25; 346 -2.5 0 1.25; 347 -2.75 0 1.25; 348 -3 0 1.25
60. 349 -3.25 0 1.25; 350 -3.5 0 1.25; 351 -3.75 0 1.25; 352 -4 0 1.25
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190. MEMBER INCIDENCES
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192. ELEMENT INCIDENCES SHELL
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429. START GROUP DEFINITION
430. ELEMENT

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431. _BM 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 -
432. 105 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 -
433. 171 TO 177 182 TO 188 193 TO 199 294 TO 300 302 TO 304 307 308 310 TO 312 -
434. 315 316 319 TO 325 327 TO 333 335 TO 341 343 TO 345 348 349 351 TO 353 356 -
435. 357 359 TO 365 367 TO 373 375 TO 401 433 TO 440 443 TO 450 453 TO 702 705 -
436. 706 TO 707 710 TO 731 734 TO 736 739 TO 933 935 TO 965
437. _SF 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 -
438. 105 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 -
439. 294 TO 300 375 TO 401 440 450 453 TO 611 615 627 TO 640 644 656 TO 669 673 -
440. 685 TO 698 702 714 TO 727 731 743 TO 756 760 772 TO 785 789 830 TO 843 847 -
441. 859 TO 872 876 895 TO 900 918 TO 923 935 TO 948 952
442. END GROUP DEFINITION
443. ELEMENT PROPERTY
444. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
445. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
446. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
447. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
448. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 THICKNESS 0.65
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453. DENSITY 2.40262
454. ALPHA 1E-005
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457. STRENGTH FCU 2812.28
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462. MATERIAL CONCRETE ALL
463. SUPPORTS
464. 1 2 5 9 TO 16 18 TO 25 27 TO 34 36 TO 43 45 TO 52 54 TO 61 63 TO 70 -
465. 72 TO 79 81 TO 88 90 TO 97 99 TO 106 108 TO 115 117 TO 124 126 TO 133 135 -
466. 136 TO 142 144 TO 151 153 TO 160 162 TO 167 246 TO 253 255 TO 258 260 TO 262 -
467. 264 TO 271 273 TO 281 283 TO 290 292 TO 299 301 TO 307 309 TO 316 -
468. 318 TO 325 327 TO 334 336 TO 363 392 TO 399 401 403 TO 410 412 414 416 418 -
469. 419 TO 945 ELASTIC MAT DIRECT Y SUBGRADE 4500 PRINT
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941	0.023437
942	0.007812

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       943          0.015625
       944          0.015625
       945          0.007813
    TOTAL          42.593750
470. LOAD 1 DL+LL
471. SELFWEIGHT Y -1
472. ELEMENT LOAD
473. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
474. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
475. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
476. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
477. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
478. JOINT LOAD
479. 3 FX 0.66 FY -32.4 FZ -0.13
480. 415 FX 0.14 FY -51.24 FZ 0.09
481. 4 FX 0.9 FY -29.18 FZ 0.36
482. 417 FX 0.4 FY -38.56 FZ 0.55
483. LOAD 2 DL+0.75*LL+0.75*WLX
484. SELFWEIGHT Y -1
485. ELEMENT LOAD
486. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
487. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
488. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
489. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
490. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
491. JOINT LOAD
492. 3 FX 7.04 FY -21.81 FZ -0.45
493. 415 FX 2.54 FY -50.48 FZ -0.23
494. 4 FX 7.3 FY -57.98 FZ 1.25
495. 417 FX 2.81 FY -49.59 FZ 1.5
496. LOAD 3 DL+0.75*LL+0.75*WLZ
497. SELFWEIGHT Y -1
498. ELEMENT LOAD
499. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
500. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
501. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
502. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
503. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.6
504. JOINT LOAD
505. 3 FX 1.46 FY -19.58 FZ 3.71
506. 415 FX 0.65 FY -77.96 FZ 4.09
507. 4 FX 1.7 FY -24.72 FZ 3.43
508. 417 FX 0.93 FY -57.97 FZ 3.78
509. LOAD 4 0.6*DL+WLX
510. SELFWEIGHT Y -1
511. ELEMENT LOAD
512. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
513. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
514. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
515. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
516. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3
517. JOINT LOAD
518. 3 FX 7.6 FY 20.12 FZ -0.45
519. 415 FX 2.95 FY -2.05 FZ -0.42
520. 4 FX 7.74 FY -32.16 FZ 0.8
521. 417 FX 3.09 FY -18.91 FZ 0.92
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522. LOAD 5 0.6*DL+WLZ
523. SELFWEIGHT Y -1
524. ELEMENT LOAD
525. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
526. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
527. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
528. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
529. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3
530. JOINT LOAD
531. 3 FX 0.16 FY 23.09 FZ 5.1
532. 415 FX 0.42 FY -38.7 FZ 5.34
533. 4 FX 0.27 FY 12.19 FZ 3.7
534. 417 FX 0.58 FY -30.08 FZ 3.95
535. ******************************************
536. LOAD 100 1.4*DL
537. SELFWEIGHT Y -1
538. ELEMENT LOAD
539. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
540. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
541. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
542. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
543. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4.3
544. JOINT LOAD
545. 3 FX -0.19 FY -17.95 FZ -0.09
546. 415 FX 0.11 FY -24.46 FZ -0.09
547. 4 FX 0.11 FY -15.55 FZ -0.03
548. 417 FX 0.2 FY -19.05 FZ 0.14
549. LOAD 101 1.4*DL + 1.7*LL
550. SELFWEIGHT Y -1
551. ELEMENT LOAD
552. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
553. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
554. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
555. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
556. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4.3
557. JOINT LOAD
558. 3 FX 1.16 FY -51.23 FZ -0.2
559. 415 FX 0.27 FY -81.87 FZ 0.14
560. 4 FX 1.5 FY -46.27 FZ 0.62
561. 417 FX 0.64 FY -61.47 FZ 0.91
562. LOAD 102 1.2*DL + 1.6*WLX + LL
563. SELFWEIGHT Y -1
564. ELEMENT LOAD
565. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
566. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
567. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
568. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
569. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4
570. JOINT LOAD
571. 3 FX 12.91 FY 9.54 FZ -0.8
572. 415 FX 4.93 FY -41.24 FZ -0.62
573. 4 FX 13.22 FY -72.19 FZ 1.65
574. 417 FX 5.24 FY -58.48 FZ 1.95
575. LOAD 103 1.2*DL + 1.6*WLZ + LL
576. SELFWEIGHT Y -1
577. ELEMENT LOAD
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578. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
579. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
580. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
581. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
582. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -4
583. JOINT LOAD
584. 3 FX 1.01 FY 14.3 FZ 8.08
585. 415 FX 0.88 FY -99.88 FZ 8.6
586. 4 FX 1.26 FY -1.23 FZ 6.3
587. 417 FX 1.22 FY -76.36 FZ 6.81
588. LOAD 104 0.9*DL + 1.6*WLX
589. SELFWEIGHT Y -1
590. ELEMENT LOAD
591. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
592. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
593. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
594. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
595. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.5
596. JOINT LOAD
597. 3 FX 12.16 FY 32.96 FZ -0.72
598. 415 FX 4.73 FY -2.23 FZ -0.67
599. 4 FX 12.37 FY -50.79 FZ 1.27
600. 417 FX 4.94 FY -29.44 FZ 1.46
601. LOAD 105 0.9*DL + 1.6*WLZ
602. SELFWEIGHT Y -1
603. ELEMENT LOAD
604. 17 TO 23 28 TO 34 39 TO 45 50 TO 56 61 TO 67 72 TO 78 83 TO 89 94 TO 100 105 -
605. 106 TO 111 116 TO 122 127 TO 133 138 TO 144 149 TO 155 160 TO 166 171 TO 177 -
606. 182 TO 188 193 TO 199 294 TO 300 302 TO 308 310 TO 316 319 TO 325 -
607. 327 TO 333 335 TO 341 343 TO 349 351 TO 357 359 TO 365 367 TO 373 -
608. 375 TO 401 433 TO 440 443 TO 450 453 TO 933 935 TO 965 PR -3.5
609. JOINT LOAD
610. 3 FX 0.26 FY 37.72 FZ 8.16
611. 415 FX 0.68 FY -60.87 FZ 8.55
612. 4 FX 0.42 FY 20.17 FZ 5.92
613. 417 FX 0.92 FY -47.32 FZ -6.32
614. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**804**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**744**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**800**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 612/ 41/ 126 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 2424
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 306 DOUBLE KILO-WORDS

STAAD SPACE **-- PAGE NO. 27**

REQRD/AVAIL. DISK SPACE = 19.5/ 113632.1 MB

615. START CONCRETE DESIGN

```
616. CODE ACI
617. CLS 0.0075 ALL
618. TRACK 0 ALL
619. DESIGN COLUMN 429 430 451 452
```

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COLUMN NO. 429 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 430 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 451 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==========

COLUMN NO. 452 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 16 MM	1.001	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

620. END CONCRETE DESIGN
621. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 11:26: 2 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revision No.:	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 5 STAAD Input & Output File of Foundation Mkd. F7

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 22, 2020                *
*           Time=    15:50:49                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDN_F7.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 06-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 3 0 2 0; 9 -0.75 0 0; 10 -0.5 0 0; 11 -0.25 0 0; 12 0.25 0 0
     9. 13 0.5 0 0; 14 0.75 0 0; 15 0 0 -0.25; 18 -0.75 0 -0.25; 19 -0.5 0 -0.25
    10. 20 -0.25 0 -0.25; 21 0.25 0 -0.25; 22 0.5 0 -0.25; 23 0.75 0 -0.25
    11. 24 0 0 -0.5; 27 -0.75 0 -0.5; 28 -0.5 0 -0.5; 29 -0.25 0 -0.5; 30 0.25 0 -0.5
    12. 31 0.5 0 -0.5; 32 0.75 0 -0.5; 33 0 0 -0.75; 36 -0.75 0 -0.75; 37 -0.5 0 -0.75
    13. 38 -0.25 0 -0.75; 39 0.25 0 -0.75; 40 0.5 0 -0.75; 41 0.75 0 -0.75; 60 0 0 -1.
    14. 63 -0.75 0 -1; 64 -0.5 0 -1; 65 -0.25 0 -1; 66 0.25 0 -1; 67 0.5 0 -1.
    15. 68 0.75 0 -1; 69 0 0 -1.25; 72 -0.75 0 -1.25; 73 -0.5 0 -1.25
    16. 74 -0.25 0 -1.25; 75 0.25 0 -1.25; 76 0.5 0 -1.25; 77 0.75 0 -1.25
    17. 78 0 0 -1.5; 81 -0.75 0 -1.5; 82 -0.5 0 -1.5; 83 -0.25 0 -1.5; 84 0.25 0 -1.5
    18. 85 0.5 0 -1.5; 86 0.75 0 -1.5; 87 0 0 -1.75; 90 -0.75 0 -1.75; 91 -0.5 0 -1.75
    19. 92 -0.25 0 -1.75; 93 0.25 0 -1.75; 94 0.5 0 -1.75; 95 0.75 0 -1.75
    20. 247 -0.75 0 0.25; 248 -0.5 0 0.25; 249 -0.25 0 0.25; 250 0 0 0.25
    21. 251 0.25 0 0.25; 252 0.5 0 0.25; 253 0.75 0 0.25; 256 -0.75 0 0.5
    22. 257 -0.5 0 0.5; 258 -0.25 0 0.5; 259 0 0 0.5; 260 0.25 0 0.5; 261 0.5 0 0.5
    23. 262 0.75 0 0.5; 264 -0.75 0 -2; 266 -0.5 0 -2; 267 -0.25 0 -2; 268 0 0 -2
    24. 269 0.25 0 -2; 270 0.5 0 -2; 271 0.75 0 -2; 272 0 2 -1.5
    25. MEMBER INCIDENCES
    26. 1 1 3; 334 78 272
    27. ELEMENT INCIDENCES SHELL
    28. 18 9 10 19 18; 19 10 11 20 19; 20 11 1 15 20; 21 1 12 21 15; 22 12 13 22 21
    29. 23 13 14 23 22; 29 18 19 28 27; 30 19 20 29 28; 31 20 15 24 29; 32 15 21 30 24
    30. 33 21 22 31 30; 34 22 23 32 31; 40 27 28 37 36; 41 28 29 38 37; 42 29 24 33 38
    31. 43 24 30 39 33; 44 30 31 40 39; 45 31 32 41 40; 73 36 37 64 63; 74 37 38 65 64
    32. 75 38 33 60 65; 76 33 39 66 60; 77 39 40 67 66; 78 40 41 68 67; 84 63 64 73 72
    33. 85 64 65 74 73; 86 65 60 69 74; 87 60 66 75 69; 88 66 67 76 75; 89 67 68 77 76
    34. 95 72 73 82 81; 96 73 74 83 82; 97 74 69 78 83; 98 69 75 84 78; 99 75 76 85 84
    35. 100 76 77 86 85; 106 81 82 91 90; 107 82 83 92 91; 108 83 78 87 92
    36. 109 78 84 93 87; 110 84 85 94 93; 111 85 86 95 94; 295 247 248 10 9
    37. 296 248 249 11 10; 297 249 250 1 11; 298 250 251 12 1; 299 251 252 13 12
    38. 300 252 253 14 13; 320 256 257 248 247; 321 257 258 249 248
```

```
39. 322 258 259 250 249; 323 259 260 251 250; 324 260 261 252 251
40. 325 261 262 253 252; 328 90 91 266 264; 329 91 92 267 266; 330 92 87 268 267
41. 331 87 93 269 268; 332 93 94 270 269; 333 94 95 271 270
42. START GROUP DEFINITION
43. ELEMENT
44. _BM 18 19 22 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 96 99 100 106 107 110 -
45. 111 295 296 299 300 320 TO 325 328 TO 333
46. _SF 18 23 29 34 40 TO 45 73 TO 78 84 89 95 100 106 111 295 300 320 325 328 333
47. END GROUP DEFINITION
48. ELEMENT PROPERTY
49. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
50. 320 TO 325 328 TO 333 THICKNESS 0.3
51. DEFINE MATERIAL START
52. ISOTROPIC CONCRETE
53. E 2.21467E+006
54. POISSON 0.17
55. DENSITY 2.40262
56. ALPHA 1E-005
57. DAMP 0.05
58. TYPE CONCRETE
59. STRENGTH FCU 2812.28
60. END DEFINE MATERIAL
61. MEMBER PROPERTY AMERICAN
62. 1 334 PRIS YD 0.6 ZD 0.6
63. CONSTANTS
64. MATERIAL CONCRETE ALL
65. SUPPORTS
66. 1 9 TO 15 18 TO 24 27 TO 33 36 TO 41 60 63 TO 69 72 TO 78 81 TO 87 90 TO 95 -
67. 247 TO 253 256 TO 262 264 266 TO 270 -
68. 271 ELASTIC MAT DIRECT Y SUBGRADE 4500 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
9	0.031250
10	0.062500
11	0.062500
12	0.062500
13	0.062500
14	0.031250
15	0.062500
18	0.031250
19	0.062500
20	0.062500
21	0.062500
22	0.062500
23	0.031250
24	0.062500
27	0.031250
28	0.062500
29	0.062500
30	0.062500
31	0.062500
32	0.031250
33	0.062500

36	0.031250
37	0.062500
38	0.062500
39	0.062500
40	0.062500
41	0.031250
60	0.062500
63	0.031250
64	0.062500
65	0.062500
66	0.062500
67	0.062500
68	0.031250
69	0.062500
72	0.031250
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.031250
78	0.062500
81	0.031250
82	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.031250
87	0.062500
90	0.031250
91	0.062500
92	0.062500
93	0.062500
94	0.062500
95	0.031250
247	0.031250
248	0.062500
249	0.062500
250	0.062500
251	0.062500
252	0.062500
253	0.031250
256	0.015625
257	0.031250
258	0.031250
259	0.031250
260	0.031250
261	0.031250
262	0.015625
264	0.015625
266	0.031250
267	0.031250
268	0.031250
269	0.031250
270	0.031250
271	0.015625
TOTAL	3.750000

```
LOAD 1 DL+LL
*SOIL+SURCHARGE=1.7X1.8+2=5T/SQ.M)
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -4.2
JOINT LOAD
3 FX 0.01 FY -16 FZ 1
272 FX 0.01 FY -11.5 FZ -0.2
**************************************************************************
LOAD 2 DL+0.75*LL+0.75*WLX
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -4.2
JOINT LOAD
3 FX 0.1 FY -13.5 FZ 1
272 FX 0.1 FY -10 FZ -0.2
**************************************************************************
LOAD 3 DL+0.75*LL+0.75*WLZ
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -4.2
JOINT LOAD
3 FX 0.01 FY -20 FZ 2
272 FX 0.01 FY -4 FZ 1
**************************************************************************
LOAD 4 0.6*DL+WLX
*SOIL+SURCHARGE=1.7X1.8+0.6*2=4.3T/SQ.M
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -3.5
JOINT LOAD
3 FX 0.1 FY -3 FZ 0.2
272 FX 0.1 FY -3 FZ -0.2
**************************************************************************
LOAD 5 0.6*DL+WLZ
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -3.5
JOINT LOAD
3 FX 0.01 FY -11 FZ 2
272 FX 0.01 FY 5 FZ 2
**************************************************************************
**********************************************
LOAD 100 1.4*DL
*SOIL+SURCHARGE=1.7X1.8+1.4*2=5.86T/SQ.M
SELFWEIGHT Y -1
ELEMENT LOAD
18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
320 TO 325 328 TO 333 PR -5.9
JOINT LOAD
3 272 FX 0.01 FY -7.41 FZ 0
```

```
125. ************************************8
126. LOAD 101 1.4*DL + 1.7*LL
127. SELFWEIGHT Y -1
128. ELEMENT LOAD
129. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
130. 320 TO 325 328 TO 333 PR -5
131. JOINT LOAD
132. 3 FX 0.01 FY -25 FZ 1.75
133. 272 FX 0.01 FY -17.5 FZ -0.22
134. ************************************
135. LOAD 102 1.2*DL + 1.6*WLX + LL
136. *SOIL+SURCHARGE=1.2X1.8+1.2*2=5.0T/SQ.M
137. ELEMENT LOAD
138. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
139. 320 TO 325 328 TO 333 PR -5
140. JOINT LOAD
141. 3 FX 0.1 FY -17 FZ 1.12
142. 272 FX 0.1 FY -12.5 FZ -0.22
143. ********************************
144. LOAD 103 1.2*DL + 1.6*WLZ + LL
145. ELEMENT LOAD
146. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
147. 320 TO 325 328 TO 333 PR -5
148. JOINT LOAD
149. 3 FX -0.01 FY -30 FZ 4
150. 272 FX -0.01 FY 0.5 FZ 3
151. ************************************
152. LOAD 104 0.9*DL + 1.6*WLX
153. *SOIL+SURCHARGE=1.2X1.8+0.9*2=4T/SQ.M
154. ELEMENT LOAD
155. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
156. 320 TO 325 328 TO 333 PR -4
157. JOINT LOAD
158. 3 FX 0.1 FY -5 FZ 0.3
159. 272 FX 0.1 FY -5 FZ -0.3
160. ************************************8
161. LOAD 105 0.9*DL + 1.6*WLZ
162. ELEMENT LOAD
163. 18 TO 23 29 TO 34 40 TO 45 73 TO 78 84 TO 89 95 TO 100 106 TO 111 295 TO 300 -
164. 320 TO 325 328 TO 333 PR -4
165. JOINT LOAD
166. 3 FX -0.01 FY -17.5 FZ 3
167. 272 FX -0.01 FY 8 FZ 3
168. ********************************
169. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**79**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**60**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**77**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 61/ 12/ 39 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 243
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 10 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 113604.1 MB

170. START CONCRETE DESIGN

```
171. CODE ACI
172. CLS 0.075 ALL
173. DESIGN COLUMN 1 334
```

==

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 334 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

174. END CONCRETE DESIGN
175. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 15:51: 1 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	0			Outotec / L&T Revision No.:	0	1	2	3	
Date	22-05-2020	29-06-2020	16.11.2023				Date	22-05-2020	29-06-2020	16.11.2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 6 STAAD Input & Output File of Foundation Mkd. F9

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 22, 2020                *
*           Time=    15:52:42                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F9.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 07-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 1.5; 3 1 0 0; 4 1 0 1.5; 5 0.25 0 0; 6 0.5 0 0; 7 0.75 0 0
     9. 8 0 0 0.25; 9 0 0 0.5; 10 0 0 0.75; 11 0 0 1; 12 0 0 1.25; 13 0.25 0 0.25
    10. 14 0.5 0 0.25; 15 0.75 0 0.25; 16 1 0 0.25; 17 1.25 0 0; 18 1.25 0 0.25
    11. 19 1.5 0 0; 20 1.5 0 0.25; 21 0.25 0 0.5; 22 0.5 0 0.5; 23 0.75 0 0.5
    12. 24 1 0 0.5; 25 1.25 0 0.5; 26 1.5 0 0.5; 27 0.25 0 0.75; 28 0.5 0 0.75
    13. 29 0.75 0 0.75; 30 1 0 0.75; 31 1.25 0 0.75; 32 1.5 0 0.75; 33 0.25 0 1
    14. 34 0.5 0 1; 35 0.75 0 1; 36 1 0 1; 37 1.25 0 1; 38 1.5 0 1; 39 0.25 0 1.25
    15. 40 0.5 0 1.25; 41 0.75 0 1.25; 42 1 0 1.25; 43 1.25 0 1.25; 44 1.5 0 1.25
    16. 45 0.25 0 1.5; 46 0.5 0 1.5; 47 0.75 0 1.5; 48 1.25 0 1.5; 49 1.5 0 1.5
    17. 50 0 0 1.75; 51 0.25 0 1.75; 52 0.5 0 1.75; 53 0.75 0 1.75; 54 1 0 1.75
    18. 55 1.25 0 1.75; 56 1.5 0 1.75; 57 0 0 2; 58 0.25 0 2; 59 0.5 0 2; 60 0.75 0 2
    19. 61 1 0 2; 62 1.25 0 2; 63 1.5 0 2; 64 0 0 2.25; 65 0.25 0 2.25; 66 0.5 0 2.25
    20. 67 0.75 0 2.25; 68 1 0 2.25; 69 1.25 0 2.25; 70 1.5 0 2.25; 71 0 2.5 0
    21. 72 0 2.5 1.5; 73 1 2.5 0; 74 1 2.5 1.5; 75 -0.25 0 0; 76 -0.25 0 0.25
    22. 77 -0.25 0 0.5; 78 -0.25 0 0.75; 79 -0.25 0 1; 80 -0.25 0 1.25; 81 -0.25 0 1.5
    23. 82 -0.25 0 1.75; 83 -0.25 0 2; 84 -0.25 0 2.25; 85 -0.5 0 0; 86 -0.5 0 0.25
    24. 87 -0.5 0 0.5; 88 -0.5 0 0.75; 89 -0.5 0 1; 90 -0.5 0 1.25; 91 -0.5 0 1.5
    25. 92 -0.5 0 1.75; 93 -0.5 0 2; 94 -0.5 0 2.25; 95 0.25 0 -0.25; 96 0 0 -0.25
    26. 97 0.5 0 -0.25; 98 0.75 0 -0.25; 99 1 0 -0.25; 100 1.25 0 -0.25
    27. 101 1.5 0 -0.25; 102 -0.25 0 -0.25; 103 -0.5 0 -0.25; 104 0.25 0 -0.5
    28. 105 0 0 -0.5; 106 0.5 0 -0.5; 107 0.75 0 -0.5; 108 1 0 -0.5; 109 1.25 0 -0.5
    29. 110 1.5 0 -0.5; 111 -0.25 0 -0.5; 112 -0.5 0 -0.5; 113 0.25 0 -0.75
    30. 114 0 0 -0.75; 115 0.5 0 -0.75; 116 0.75 0 -0.75; 117 1 0 -0.75
    31. 118 1.25 0 -0.75; 119 1.5 0 -0.75; 120 -0.25 0 -0.75; 121 -0.5 0 -0.75
    32. 122 0.25 0 -1; 123 0 0 -1; 124 0.5 0 -1; 125 0.75 0 -1; 126 1 0 -1.
    33. 127 1.25 0 -1; 128 1.5 0 -1; 129 -0.25 0 -1; 130 -0.5 0 -1; 131 -0.75 0 0
    34. 132 -0.75 0 0.25; 133 -0.75 0 0.5; 134 -0.75 0 0.75; 135 -0.75 0 1
    35. 136 -0.75 0 1.25; 137 -0.75 0 1.5; 138 -0.75 0 1.75; 139 -0.75 0 2
    36. 140 -0.75 0 2.25; 141 -0.75 0 -0.25; 142 -0.75 0 -0.5; 143 -0.75 0 -0.75
    37. 144 -0.75 0 -1; 145 1.75 0 0; 146 1.75 0 0.25; 147 1.75 0 0.5; 148 1.75 0 0.75
    38. 149 1.75 0 1; 150 1.75 0 1.25; 151 1.75 0 1.5; 152 1.75 0 1.75; 153 1.75 0 2
```

39. 154 1.75 0 2.25; 155 1.75 0 -0.25; 156 1.75 0 -0.5; 157 1.75 0 -0.75
40. 158 1.75 0 -1; 159 -1 0 0; 160 -1 0 0.25; 161 -1 0 0.5; 162 -1 0 0.75
41. 163 -1 0 1; 164 -1 0 1.25; 165 -1 0 1.5; 166 -1 0 1.75; 167 -1 0 2
42. 168 -1 0 2.25; 169 -1 0 -0.25; 170 -1 0 -0.5; 171 -1 0 -0.75; 172 -1 0 -1.
43. 173 2 0 0; 174 2 0 0.25; 175 2 0 0.5; 176 2 0 0.75; 177 2 0 1; 178 2 0 1.25
44. 179 2 0 1.5; 180 2 0 1.75; 181 2 0 2; 182 2 0 2.25; 183 2 0 -0.25
45. 184 2 0 -0.5; 185 2 0 -0.75; 186 2 0 -1; 187 -1.25 0 0; 188 -1.25 0 0.25
46. 189 -1.25 0 0.5; 190 -1.25 0 0.75; 191 -1.25 0 1; 192 -1.25 0 1.25
47. 193 -1.25 0 1.5; 194 -1.25 0 1.75; 195 -1.25 0 2; 196 -1.25 0 2.25
48. 197 -1.25 0 -0.25; 198 -1.25 0 -0.5; 199 -1.25 0 -0.75; 200 -1.25 0 -1.
49. 201 2.25 0 0; 202 2.25 0 0.25; 203 2.25 0 0.5; 204 2.25 0 0.75; 205 2.25 0 1
50. 206 2.25 0 1.25; 207 2.25 0 1.5; 208 2.25 0 1.75; 209 2.25 0 2
51. 210 2.25 0 2.25; 211 2.25 0 -0.25; 212 2.25 0 -0.5; 213 2.25 0 -0.75
52. 214 2.25 0 -1; 215 0 0 2.5; 216 0.25 0 2.5; 217 0.5 0 2.5; 218 0.75 0 2.5
53. 219 1 0 2.5; 220 1.25 0 2.5; 221 1.5 0 2.5; 222 -0.25 0 2.5; 223 -0.5 0 2.5
54. 224 -0.75 0 2.5; 225 1.75 0 2.5; 226 -1 0 2.5; 227 2 0 2.5; 228 -1.25 0 2.5
55. 229 2.25 0 2.5
56. MEMBER INCIDENCES
57. 65 1 71; 66 2 72; 67 3 73; 68 4 74
58. ELEMENT INCIDENCES SHELL
59. 11 5 1 8 13; 12 6 5 13 14; 13 7 6 14 15; 14 3 7 15 16; 15 17 3 16 18
60. 16 19 17 18 20; 17 13 8 9 21; 18 14 13 21 22; 19 15 14 22 23; 20 16 15 23 24
61. 21 18 16 24 25; 22 20 18 25 26; 23 21 9 10 27; 24 22 21 27 28; 25 23 22 28 29
62. 26 24 23 29 30; 27 25 24 30 31; 28 26 25 31 32; 29 27 10 11 33; 30 28 27 33 34
63. 31 29 28 34 35; 32 30 29 35 36; 33 31 30 36 37; 34 32 31 37 38; 35 33 11 12 39
64. 36 34 33 39 40; 37 35 34 40 41; 38 36 35 41 42; 39 37 36 42 43; 40 38 37 43 44
65. 41 39 12 2 45; 42 40 39 45 46; 43 41 40 46 47; 44 42 41 47 4; 45 43 42 4 48
66. 46 44 43 48 49; 47 45 2 50 51; 48 46 45 51 52; 49 47 46 52 53; 50 4 47 53 54
67. 51 48 4 54 55; 52 49 48 55 56; 53 51 50 57 58; 54 52 51 58 59; 55 53 52 59 60
68. 56 54 53 60 61; 57 55 54 61 62; 58 56 55 62 63; 59 58 57 64 65; 60 59 58 65 66
69. 61 60 59 66 67; 62 61 60 67 68; 63 62 61 68 69; 64 63 62 69 70; 69 1 75 76 8
70. 70 8 76 77 9; 71 9 77 78 10; 72 10 78 79 11; 73 11 79 80 12; 74 12 80 81 2
71. 75 2 81 82 50; 76 50 82 83 57; 77 57 83 84 64; 78 75 85 86 76; 79 76 86 87 77
72. 80 77 87 88 78; 81 78 88 89 79; 82 79 89 90 80; 83 80 90 91 81; 84 81 91 92 82
73. 85 82 92 93 83; 86 83 93 94 84; 87 95 96 1 5; 88 97 95 5 6; 89 98 97 6 7
74. 90 99 98 7 3; 91 100 99 3 17; 92 101 100 17 19; 93 96 102 75 1
75. 94 102 103 85 75; 95 104 105 96 95; 96 106 104 95 97; 97 107 106 97 98
76. 98 108 107 98 99; 99 109 108 99 100; 100 110 109 100 101; 101 105 111 102 96
77. 102 111 112 103 102; 103 113 114 105 104; 104 115 113 104 106
78. 105 116 115 106 107; 106 117 116 107 108; 107 118 117 108 109
79. 108 119 118 109 110; 109 114 120 111 105; 110 120 121 112 111
80. 111 122 123 114 113; 112 124 122 113 115; 113 125 124 115 116
81. 114 126 125 116 117; 115 127 126 117 118; 116 128 127 118 119
82. 117 123 129 120 114; 118 129 130 121 120; 119 85 131 132 86; 120 86 132 133 87
83. 121 87 133 134 88; 122 88 134 135 89; 123 89 135 136 90; 124 90 136 137 91
84. 125 91 137 138 92; 126 92 138 139 93; 127 93 139 140 94; 128 103 141 131 85
85. 129 112 142 141 103; 130 121 143 142 112; 131 130 144 143 121
86. 132 145 19 20 146; 133 146 20 26 147; 134 147 26 32 148; 135 148 32 38 149
87. 136 149 38 44 150; 137 150 44 49 151; 138 151 49 56 152; 139 152 56 63 153
88. 140 153 63 70 154; 141 155 101 19 145; 142 156 110 101 155
89. 143 157 119 110 156; 144 158 128 119 157; 145 131 159 160 132
90. 146 132 160 161 133; 147 133 161 162 134; 148 134 162 163 135
91. 149 135 163 164 136; 150 136 164 165 137; 151 137 165 166 138
92. 152 138 166 167 139; 153 139 167 168 140; 154 141 169 159 131
93. 155 142 170 169 141; 156 143 171 170 142; 157 144 172 171 143
94. 158 173 145 146 174; 159 174 146 147 175; 160 175 147 148 176

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 95. 161 176 148 149 177; 162 177 149 150 178; 163 178 150 151 179
 96. 164 179 151 152 180; 165 180 152 153 181; 166 181 153 154 182
 97. 167 183 155 145 173; 168 184 156 155 183; 169 185 157 156 184
 98. 170 186 158 157 185; 171 159 187 188 160; 172 160 188 189 161
 99. 173 161 189 190 162; 174 162 190 191 163; 175 163 191 192 164
100. 176 164 192 193 165; 177 165 193 194 166; 178 166 194 195 167
101. 179 167 195 196 168; 180 169 197 187 159; 181 170 198 197 169
102. 182 171 199 198 170; 183 172 200 199 171; 184 201 173 174 202
103. 185 202 174 175 203; 186 203 175 176 204; 187 204 176 177 205
104. 188 205 177 178 206; 189 206 178 179 207; 190 207 179 180 208
105. 191 208 180 181 209; 192 209 181 182 210; 193 211 183 173 201
106. 194 212 184 183 211; 195 213 185 184 212; 196 214 186 185 213
107. 197 65 64 215 216; 198 66 65 216 217; 199 67 66 217 218; 200 68 67 218 219
108. 201 69 68 219 220; 202 70 69 220 221; 203 64 84 222 215; 204 84 94 223 222
109. 205 94 140 224 223; 206 154 70 221 225; 207 140 168 226 224
110. 208 182 154 225 227; 209 168 196 228 226; 210 210 182 227 229
111. START GROUP DEFINITION
112. ELEMENT
113. _BM 12 13 16 TO 40 42 43 46 48 49 52 TO 64 70 TO 73 76 TO 86 88 89 92 94 TO 210
114. _SF 23 TO 34 59 TO 64 71 72 77 80 81 86 103 TO 210
115. END GROUP DEFINITION
116. ELEMENT PROPERTY
117. 11 TO 64 69 TO 210 THICKNESS 0.3
118. DEFINE MATERIAL START
119. ISOTROPIC CONCRETE
120. E 2.21467E+006
121. POISSON 0.17
122. DENSITY 2.40262
123. ALPHA 1E-005
124. DAMP 0.05
125. TYPE CONCRETE
126. STRENGTH FCU 2812.28
127. END DEFINE MATERIAL
128. MEMBER PROPERTY AMERICAN
129. 65 TO 68 PRIS YD 0.6 ZD 0.6
130. CONSTANTS
131. MATERIAL CONCRETE ALL
132. SUPPORTS
133. 1 TO 70 75 TO 229 ELASTIC MAT DIRECT Y SUBGRADE 4800 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
3	0.062500
4	0.062500
5	0.062500
6	0.062500
7	0.062500
8	0.062500
9	0.062500
10	0.062500
11	0.062500
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115	0.062500
116	0.062500
117	0.062500
118	0.062500
119	0.062500
120	0.062500
121	0.062500
122	0.031250
123	0.031250
124	0.031250
125	0.031250
126	0.031250
127	0.031250
128	0.031250
129	0.031250

130	0.031250
131	0.062500
132	0.062500
133	0.062500
134	0.062500
135	0.062500
136	0.062500
137	0.062500
138	0.062500
139	0.062500
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141	0.062500
142	0.062500
143	0.062500
144	0.031250
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062500
151	0.062500
152	0.062500
153	0.062500
154	0.062500
155	0.062500
156	0.062500
157	0.062500
158	0.031250
159	0.062500
160	0.062500
161	0.062500
162	0.062500
163	0.062500
164	0.062500
165	0.062500
166	0.062500
167	0.062500
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171	0.062500
172	0.031250
173	0.062500
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179	0.062500
180	0.062500
181	0.062500
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184	0.062500
185	0.062500

186	0.031250
187	0.031250
188	0.031250
189	0.031250
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191	0.031250
192	0.031250
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194	0.031250
195	0.031250
196	0.031250
197	0.031250
198	0.031250
199	0.031250
200	0.015625
201	0.031250
202	0.031250
203	0.031250
204	0.031250
205	0.031250
206	0.031250
207	0.031250
208	0.031250
209	0.031250
210	0.031250
211	0.031250
212	0.031250
213	0.031250
214	0.015625
215	0.031250
216	0.031250
217	0.031250
218	0.031250
219	0.031250
220	0.031250
221	0.031250
222	0.031250
223	0.031250
224	0.031250
225	0.031250
226	0.031250
227	0.031250
228	0.015625
229	0.015625
TOTAL	12.250000

```
134. LOAD 1 LOADTYPE NONE  TITLE LOAD CASE 1
135. SELFWEIGHT Y -1
136. ELEMENT LOAD
137. 11 TO 64 69 TO 210 PR GY -6
138. JOINT LOAD
139. 73 74 FX -2.15 FY 1.6
140. 71 72 FX -2.15 FY -1.8
141. LOAD COMB 100 1.4DL
142. 1 1.4
143. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 229 NUMBER OF MEMBERS 4
NUMBER OF PLATES 196 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 225

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 158/ 25/ 81 DOF
TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 699
TOTAL LOAD COMBINATION CASES = 1 SO FAR.
SIZE OF STIFFNESS MATRIX = 57 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.6/ 113603.2 MB

144. **********88
145. LOAD LIST 100
146. START CONCRETE DESIGN

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147. CODE ACI 1999
148. TRACK 0 ALL
149. MAXMAIN 25 ALL
150. CLB 0.075 MEMB 65 TO 68
151. CLS 0.075 ALL
152. DESIGN COLUMN 65 TO 68
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==

COLUMN NO. 65 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 66 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 67 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 68 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

153. END CONCRETE DESIGN
154. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 15:52:51 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
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